

Kernel Density Estimation by Spectral Decomposition: Data-Driven Tapering and Superposition

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Abstract—Kernel density estimation is among the most widely used tools in nonparametric statistics, and its accuracy depends largely on one choice, the smoothing bandwidth. This paper treats bandwidth selection and density estimation in the characteristic-function domain, where the cyclic group-averaged covariance of the binned data has the squared empirical characteristic function as its spectrum: the true characteristic function sits over a sampling-noise floor of $1/n$, and the bandwidth is the spectral cutoff where the two meet. From this reading follow several methods. An automatic bandwidth selector strips the floor and minimizes a frequency-domain error criterion, matching the rule of thumb on smooth densities and approaching the best fixed bandwidth on multimodal ones, where the rule of thumb over-smooths. An adaptive estimator generalizes the fixed kernel to the per-frequency optimal Wiener taper, matching or surpassing the best fixed-bandwidth kernel on most standard densities, including sharply peaked and comb-like cases on which fixed-bandwidth methods fail; deconvolution under known measurement error follows in the same domain by dividing out the noise characteristic function. Because the Wiener estimator resolves sharp structure but does not fit smooth bases as economically as a mixture, a Gaussian mixture is combined with it two ways, a piecewise partition and a superposition of a smooth base and a band-limited residual, of which the superposition is made the default. A data-driven floor read from the spectrum replaces the assumed $1/n$ floor and stays robust on heaped and rounded data, where aliasing lifts the true floor. On the Marron-Wand benchmark scored by exact integrated squared error, the advantage emerges with sample size through a bias-variance tradeoff: the spectral estimators carry low bias but pay in variance, so a cross-validation bandwidth leads at $n = 100$ while the Wiener filter and the superposition take the top two ranks at $n = 5000$. The methods are validated on six real datasets, CRSP returns, NHANES self-reports, CMS dimuon and SDSS redshift spectra, a verified random-beacon stream, and UNSW-NB15 traffic, and on a synthetic-data quality-checking use-case. All claims are validated numerically and the experiments are reproducible.

Index Terms—Kernel density estimation, bandwidth selection, empirical characteristic function, group-averaged covariance, Wiener filter, density decomposition, deconvolution, heaped data.

I. INTRODUCTION

KERNEL density estimation (KDE) turns a sample into a smooth estimate of its probability density and is one of the most heavily used methods in applied statistics and machine learning [1], [2], [3]. It underlies anomaly and novelty detection, where low estimated density flags outliers,

and in cybersecurity the same principle drives behavioral baselining and zero-trust monitoring, where deviations from a learned density of normal activity raise alerts. It supports generative and probabilistic machine learning, where densities are sampled, compared, and used to score representations, including emerging uses on latent and embedding spaces for novelty detection, dataset-quality assessment, and large-language-model hallucination detection, settings in which the geometry of the embedding space may itself carry symmetry that an algebraic estimator can exploit. In quantitative finance and risk it builds the return and loss densities behind value-at-risk, expected shortfall, and tail-risk analysis, where the distributions are routinely skewed, heavy-tailed, and multimodal, a regime that motivates the attention this paper gives to asymmetric and sharply peaked targets. In the life sciences it estimates gene-expression distributions, cell-population structure, and protein-abundance profiles; in geographic information systems it smooths incidence surfaces for crime analysis, traffic-accident mapping, and disease-outbreak surveillance; and in autonomous systems and robotics it models obstacle and pedestrian densities, vehicle-movement patterns, and the likelihood fields used in localization. In signal processing it estimates spectrum-occupancy, emitter, interference, and radar-target distributions; in computer vision it models color, texture, and object-location distributions for tracking and for intensity-based image segmentation without assuming a Gaussian mixture; and in the sciences it smooths source and clustering densities in astronomy, firing-rate densities in neuroscience, and incidence surfaces in epidemiology. Across all of these the estimate is only as good as its smoothing.

That smoothing is set by the bandwidth, and some practitioners consider its selection to be the most challenging issue in kernel density estimation [3], [4]. Too small a bandwidth produces a spiky, high-variance estimate; too large a bandwidth blurs real structure into a featureless mound. The standard rules of thumb assume a nearly Gaussian target and over-smooth anything multimodal, while the more elaborate data-driven selectors trade that bias for instability or for a delicate optimization.

This paper approaches the problem through algebraic diversity [5], whose central object is the group-averaged covariance, a projection of the sample covariance onto the commutant of a group representation. Applied to the binned sample, that projection has a familiar spectrum: its eigenvalues are the squared empirical characteristic function. A kernel estimate is a low-pass filter on that same characteristic function, so the

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bandwidth is a spectral cutoff, and selecting it is the spectral-support problem of finding where the characteristic function descends into its $1/n$ sampling-noise floor. Two enhancements follow. The first is an automatic bandwidth selector built from the noise-floor strip of the algebraic framework. The second goes beyond bandwidth selection entirely: replacing the fixed kernel with the per-frequency optimal taper yields an adaptive estimator that outperforms the best fixed-bandwidth kernel.

The contributions are the following. The KDE bandwidth is identified with the spectral support of the binned data, recasting selection as the problem of locating where the empirical characteristic function meets its sampling-noise floor, and from this reading follow the bandwidth selector and the adaptive Wiener estimator. The noise floor is itself a contribution: the algebraic residue, the part of the spectrum that the matched group leaves unexplained, furnishes a data-driven maximum-entropy floor that supersedes the fixed $1/n$ floor and is far more robust on heaped or rounded data, where aliasing lifts the true floor above $1/n$. The Wiener estimator extends in the same domain to deconvolution under known measurement error, dividing out the noise characteristic function with the residual-meets-floor rule setting the cutoff. Because the Wiener filter resolves sharp structure but does not fit a smooth base as economically as a mixture, a Gaussian mixture is combined with it two ways, a piecewise partition and a superposition of a smooth base and a band-limited residual; the superposition decomposes densities whose sharp and smooth parts overlap, which no single bandwidth resolves, and is made the default. Both delivered densities, the band-limited adaptive estimate and the Gaussian mixture, are differentiable almost everywhere, so model builders and digital-twin constructions can optimize against the recovered density by gradient descent even when it fits sharp, spiky structure, and a synthetic-data quality-checking use-case shows the method recovering contrived target densities exactly. On the standard Marron-Wand benchmark of fifteen densities, scored by exact integrated squared error against the known truth at three sample sizes, the advantage emerges with sample size through a bias-variance tradeoff: the spectral estimators carry low bias but pay in variance, so at $n = 100$ a cross-validation bandwidth leads and AD-Wiener alone is the robust spectral choice, while at $n = 5000$ the Wiener filter and the superposition take the top two ranks among seven, ahead of every classical baseline including the Botev diffusion bandwidth, the superposition beating a Gaussian mixture even where the benchmark hands the mixture the correct model class. The reading is validated on six real datasets from distinct domains: across thirteen CRSP series the spectral and mixture estimators recover the tail risk that a Gaussian fit understates by more than two percent of a daily return; on NHANES survey self-reports the data-driven floor stays robust where rounding defeats the fixed floor; on the CMS dimuon and SDSS galaxy-redshift spectra, where sharp features carry probability mass on a smooth background, the adaptive estimator improves held-out likelihood over the ordinary kernel estimate; on a NIST randomness-beacon stream, verified uncorrupted by the statistical test suite, AD-Wiener recovers the uniform target almost exactly where a Gaussian mixture cannot; and on UNSW-NB15 network traffic

the superposition gives the lowest held-out likelihood across flow features. All claims are validated numerically and the experiments are reproducible.

II. BACKGROUND AND RELATED WORK

A. Kernel density estimation

Given samples $x_1, \dots, x_n$, the kernel density estimate with kernel K and bandwidth h is $\hat{f}_h(x) = \frac{1}{n} \sum_j K_h(x - x_j)$, $K_h(u) = h^{-1}K(u/h)$. Its mean integrated squared error decomposes into a squared bias that grows with h and a variance that falls with h , so a single scalar trades the two off [3]. The choice of kernel matters little; the choice of h is decisive.

B. Bandwidth selection

The simplest selectors are reference rules that set h from the sample spread under a Gaussian assumption, such as Silverman's and Scott's rules [1], [2]. These are fast but over-smooth structured densities. Data-driven selectors improve on them: least-squares and likelihood cross-validation; the plug-in method of Sheather and Jones, long the reference standard [4]; the diffusion estimator of Botev and colleagues, an FFT-based selector that also yields an improved Sheather-Jones bandwidth [6]; and the empirical-characteristic-function selectors of Chiu [7], which truncate the characteristic function at the frequency where it meets its sampling floor. Bandwidth selection remains the aspect of kernel density estimation that most determines accuracy and the one most studied.

C. Other enhancements

Beyond the global bandwidth, several lines of work enhance the estimator itself. Variable and adaptive bandwidth methods let the smoothing vary with local density, narrowing in dense regions and widening in sparse tails [8], [9]. Boundary-corrected estimators remove the bias that ordinary KDE suffers near the edge of a bounded support, by reflection or by boundary kernels [10], [11]. Transformation methods map the data to a scale where a single bandwidth suffices and then map back [12]. Deconvolution estimators recover a density from noisy observations by dividing out the noise characteristic function [13]. The present work adds to this literature a characteristic-function reading that yields both a bandwidth selector and, going beyond a fixed kernel, an optimal adaptive estimator, and it extends in the same domain to deconvolution under known measurement error (Section IX).

III. THE GROUP-AVERAGED SPECTRUM AND THE EMPIRICAL CHARACTERISTIC FUNCTION

Let the samples be binned into a histogram $p \in \mathbb{R}^M$ with $\sum_m p_m = 1$ on a uniform grid of spacing Δx , viewed as a signal on the cyclic index set. Let S be the unitary cyclic shift and $G = \mathbb{Z}_M$ the group it generates. The single-observation group-averaged covariance [5] is $\mathbf{R}_G = \frac{1}{M} \sum_{g \in G} (S^g p)(S^g p)^H$, the Reynolds projection of pp^H onto the commutant of the cyclic representation. Write F for the unitary DFT and

$\hat{\varphi}(t_k) = (Fp)_k$ for the empirical characteristic function at $t_k = 2\pi k/(M\Delta x)$.

Theorem 1 (Spectrum identity). *$\mathbf{R}_G$ is circulant, is diagonalized by F , and its eigenvalues are the squared empirical characteristic function (ECF): $(F\mathbf{R}_G F^H)_{kl} = \delta_{kl} |\hat{\varphi}(t_k)|^2$.*

Proof. The shift theorem gives $FS^g F^H = D^g$ with $D = \text{diag}(e^{-j2\pi k/M})$, so $F\mathbf{R}_G F^H = \frac{1}{M} \sum_g D^g (Fp)(Fp)^H D^{-g}$. The (k, l) entry of $(Fp)(Fp)^H$ is $(Fp)_k (Fp)_l^*$ and conjugation by D^g multiplies it by $e^{-j2\pi(k-l)g/M}$; character orthogonality $\frac{1}{M} \sum_g e^{-j2\pi(k-l)g/M} = \delta_{kl}$ leaves $\delta_{kl} |(Fp)_k|^2$. Diagonal in the Fourier basis means circulant [14]. $\square$

The binned data's algebraic spectrum is therefore the squared ECF, the same kind of object the bandwidth-estimation framework treats: a true characteristic function near the origin over a sampling-noise floor.

Definition 1 (Effective dimension). *With eigenvalues $\lambda_k = |\hat{\varphi}(t_k)|^2$, the participation ratio is $D = (\sum_k \lambda_k)^2 / \sum_k \lambda_k^2 = (\text{Tr } \mathbf{R}_G)^2 / \|\mathbf{R}_G\|_F^2$, the effective number of occupied spectral modes [15].*

Proposition 1 (Noise floor). $\mathbb{E}|\hat{\varphi}(t)|^2 = |\varphi(t)|^2 + (1 - |\varphi(t)|^2)/n \rightarrow 1/n$ as $\varphi(t) \rightarrow 0$.

Proof. $\mathbb{E}\hat{\varphi}(t) = \varphi(t)$ and $\text{Var } \hat{\varphi}(t) = \frac{1}{n}(1 - |\varphi(t)|^2)$, so $\mathbb{E}|\hat{\varphi}|^2 = |\varphi|^2 + \frac{1}{n}(1 - |\varphi|^2)$. $\square$

IV. SPECTRAL BANDWIDTH SELECTION BY THE ALGEBRAIC RESIDUE

A Gaussian kernel estimate has Fourier transform $\hat{\varphi}(t)\psi(ht)$ with $\psi(s) = e^{-s^2/2}$, so the bandwidth h is a low-pass cutoff on the ECF, and choosing it is choosing where to stop trusting $\hat{\varphi}$. The algebraic framework supplies the rule: strip the known floor, $\hat{S}_k = \max(|\hat{\varphi}(t_k)|^2 - 1/n, 0)$, truncated beyond the first frequency at which the smoothed squared ECF drops into the floor (this removes the rectification residual that otherwise biases the high-frequency tail), and select

$$h_{\text{AD}} = \arg \min_h \sum_k |\hat{\varphi}(t_k)|^2 \psi(ht_k)^2 - 2 \sum_k \hat{S}_k \psi(ht_k), \quad (1)$$

which estimates the integrated squared error up to a constant. This is the empirical-characteristic-function selector of [7] read through the floor strip and effective-support cutoff of the present framework.

Proposition 2 (Inflation and the strip). *For an expected spectrum with K in-band bins at level $a = S+b$ and $M-K$ out-of-band bins at the floor b , the participation ratio of the expected spectrum is $\bar{D} = (Ka + (M-K)b)^2 / (Ka^2 + (M-K)b^2)$, which tends to K as $b/a \rightarrow 0$ and to M as $a \rightarrow b$. Subtracting a consistent estimate of b and clipping restores $\bar{D} \rightarrow K$.*

Proof. The expression is Definition 1 on the two-level spectrum; the limits are immediate, and after subtraction the out-of-band level is zero, leaving K equal nonzero values. $\square$

A. The algebraic-residue floor option

The strip above assumes the floor is exactly $1/n$. That holds for an exact i.i.d. sample, but is violated whenever the data are rounded, heaped, or otherwise discretized, conditions that are common in practice and that lift the high-frequency floor above $1/n$ through aliasing. The algebraic-diversity framework supplies a data-driven alternative through the algebraic residue [5]: the component that the best-matched group cannot concentrate is the maximum-entropy, white part of the group-averaged covariance, and its level is read directly from the spectrum rather than assumed. For the squared ECF the noise bins are exponential, so the robust floor estimate is the order statistic $\hat{b} = \text{median}_k |\hat{\varphi}(t_k)|^2 / \ln 2$, the level at which the spectrum is indistinguishable from white. The structured part is then recovered not by a hard cut but by the soft Wiener (LMMSE) gain

$$w_k = \frac{(|\hat{\varphi}(t_k)|^2 - \hat{b})_+}{|\hat{\varphi}(t_k)|^2}, \quad \hat{S}_k = w_k |\hat{\varphi}(t_k)|^2, \quad (2)$$

which tapers the floor rather than truncating it and so reduces the residual structure left by a hard cut. The hard strip of (1) is the binary limit $w_k \in \{0, 1\}$ of this same gain. Because the matched group of the binned measure is known to be cyclic, this requires only the floor-and-gain primitive of the residue construction, not the group-search or deflation machinery of the general method. We keep the $1/n$ strip as the default and expose the residue strip as an option; Section VI shows the two are equivalent on exact samples but that the residue strip is far more robust under heaping.

V. ADAPTIVE WIENER DENSITY ESTIMATION

The kernel taper $\psi(ht)$ is a one-parameter family; nothing requires the optimal frequency-domain filter to lie in it. Write a general linear estimator as the inverse transform of $g(t)\hat{\varphi}(t)$ for a real filter g .

Proposition 3 (Optimal linear filter). *The filter minimizing the mean integrated squared error is the Wiener filter*

$$g^*(t) = \frac{|\varphi(t)|^2}{|\varphi(t)|^2 + (1 - |\varphi(t)|^2)/n}, \quad (3)$$

and the fixed-bandwidth kernel estimator is the restriction $g = \psi(ht)$ of this filter to a one-parameter family.

Proof. By Parseval the mean integrated squared error is $\frac{1}{2\pi} \int \mathbb{E}|g\hat{\varphi} - \varphi|^2 dt$. Since $\mathbb{E}\hat{\varphi} = \varphi$ and $\text{Var } \hat{\varphi} = (1 - |\varphi|^2)/n$, the integrand is $(g-1)^2 |\varphi|^2 + g^2 (1 - |\varphi|^2)/n$. Differentiating in g and setting to zero gives (3). Choosing $g = \psi(ht)$ recovers the kernel estimator. $\square$

Definition 2 (AD-Wiener estimator). *Estimate $|\varphi(t)|^2$ by the stabilized, floor-stripped squared ECF $\hat{S}(t)$ and form $\hat{g}(t) = \hat{S}(t)/(\hat{S}(t) + 1/n)$. The AD-Wiener density estimate is the inverse transform of $\hat{g}(t)\hat{\varphi}(t)$, clipped to be nonnegative and renormalized.*

Because $\hat{g}$ adapts the smoothing per frequency, keeping high-frequency content exactly where the data support it and discarding it elsewhere, the estimator can resolve structure

at several scales at once, which no single bandwidth can. Proposition 3 explains why it can match or surpass the best fixed bandwidth: it optimizes over all per-frequency tapers, of which the kernel family is a slice. The residue floor of Section IV-A substitutes directly here: replacing $1/n$ by $\hat{b}$ and using the soft gain (2) is exactly the AD-Wiener filter with a data-driven floor, and is the form we recommend for discretized data.

VI. EXPERIMENTS

The selector (1) and the AD-Wiener estimator of Definition 2 are benchmarked on the Marron-Wand test densities [16], including the extreme multi-scale cases, against Silverman’s rule, the diffusion estimator of [6] applied as an estimator at its improved Sheather-Jones bandwidth, the empirical- characteristic-function plug-in selector of [7], and the adaptive variable- bandwidth estimator of [8] with a square-root pilot law. Error is mean integrated squared error over replications against the known truth; for the fixed-bandwidth methods a common Gaussian kernel is used so that only the bandwidth differs, and “best fixed” is the ISE-minimizing single bandwidth, the strongest a fixed kernel can do. The AD methods here use the default $1/n$ strip. Seeds are recorded in `DATA.md`.

Table I places the algebraic methods against the nearest competitors rather than the rule of thumb alone. On the smooth unimodal densities every reasonable method sits near the best fixed bandwidth, with Silverman or Abramson better at the smaller sample and AD-Wiener better at the larger, where it edges even the best fixed bandwidth. Chiu’s ECF selector is the strongest classical competitor and is genuinely close: it is the best data-driven method on several densities at $n = 200$, because it shares the spectral viewpoint developed here. The adaptive estimator is the best method on the kurtotic spike at $n = 2000$, where a single variable bandwidth suits one sharp peak, but it over-adapts and fails on the claw and the combs; the diffusion estimator undersmooths throughout. The algebraic methods are the most consistent performers: the AD-Wiener estimator is best on the genuinely multi-scale cases, on the claw at $n = 2000$ surpassing the best fixed bandwidth itself by about twenty percent because no single bandwidth can both resolve the narrow peaks and smooth the broad base, while the fixed-kernel AD selector is the steadier choice on the comb densities.

On exact samples the residue strip of Section IV-A reproduces the default to within Monte-Carlo error, so Table I is unchanged by the choice and is not repeated. The two diverge once the floor assumption is violated. Table II rounds each sample to one decimal, a routine real-data condition, which aliases mass into the high frequencies and lifts the true floor well above $1/n$. The fixed-floor estimator then places its cutoff wrongly: for the bandwidth selector this inflates the error several-fold, and for the Wiener estimator it is catastrophic, since the unsuppressed rounding spectrum passes through the filter and the inverse transform is overwhelmed by spurious high-frequency oscillation, with the error rising by one to three orders of magnitude. The residue floor, read from the spectrum

rather than assumed, tracks the elevated level and keeps both estimators stable. Figure 1 shows the effect directly.

Figs. 2 and 3 show why. The rule of thumb fuses the claw and comb peaks into a single mode and misses the strongly skewed spike; the Sheather-Jones estimate resolves structure but adds spurious oscillation; the AD-Wiener estimate tracks the truth, resolving a sharp spike over a broad base (kurtotic), peaks of decreasing width (asymmetric claw), and a mixture of wide and razor-thin peaks (discrete comb) at once.

VII. GENERALITY OF THE ENABLING ESTIMATOR

The bandwidth estimator that enables the above is not special to densities. The same group-averaged spectrum estimates the occupied bandwidth of a signal, where $B_{\text{eff}} = D f_s / M$ with f_s the sampling rate. It is exact for a flat band (Theorem 2); for a single look at a stochastic process it reports an effective width of about half the support, a property of the single-look periodogram that averaging removes (Propositions 4 and 5); and the same floor strip that serves the density selector controls noise inflation. Validation of the signal-bandwidth estimator across signal-to-noise ratio, a comparison with the conventional occupied bandwidth, and an extension to chirps through the metaplectic chirp rate are summarized in Figs. 4–7; the same noise-floor strip and effective-support cutoff drive both applications.

Theorem 2 (Flat-band exactness). *If the spectrum occupies K bins with equal power, $D = K$ and $B_{\text{eff}} = K f_s / M$, the true occupied bandwidth.*

Proof. With K equal eigenvalues c and the rest zero, $D = (Kc)^2 / (Kc^2) = K$. $\square$

Proposition 4 (Single stochastic look). *If the in-band squared-ECF values are independent and exponential, as for one observation of a complex Gaussian band-limited process on K bins, then $\mathbb{E}[D] \approx (K + 1)/2$.*

Proof. For independent exponential λ_k with mean μ and $\mathbb{E}\lambda_k^2 = 2\mu^2$, $\mathbb{E}(\sum \lambda)^2 = K\mu^2 + K^2\mu^2$ and $\mathbb{E}\sum \lambda^2 = 2K\mu^2$; the ratio of expectations is $(K + 1)/2$. $\square$

Proposition 5 (Consistency under averaging). *The periodogram averaged over L independent observations converges to the power spectral density, and its participation ratio converges to that of the spectrum; for a flat band $D \rightarrow K$.*

Proof. The strong law gives bin-wise convergence; D is continuous in the eigenvalues away from the origin. $\square$

VIII. RECOGNIZING AND HANDLING ASYMMETRIC AND HEAVY-TAILED DENSITIES

The densities on which a fixed kernel struggles most are the sharply peaked and the strongly asymmetric, where one global smoothing cannot serve a narrow peak and a broad tail at once. The algebraic reading both detects this regime and suggests a remedy. As a detector, the effective dimension D_2 of the stripped coherent spectrum (Definition 1) is elevated for spike-plus-tail densities: about 5.9 for the kurtotic density and 4.3 for the strongly skewed one, against 2.2 to 2.9 for the

TABLE I
MEAN INTEGRATED SQUARED ERROR ($\times 10^3$) ON EXACT SAMPLES. “BEST FIXED” IS THE ISE-MINIMIZING SINGLE BANDWIDTH (GAUSSIAN KERNEL); THE BEST DATA-DRIVEN METHOD IS IN **BOLD**. BOTEV IS THE DIFFUSION ESTIMATOR AT THE IMPROVED SHEATHER-JONES BANDWIDTH [6], CHIU THE ECF PLUG-IN SELECTOR [7], ABRAM. THE ADAPTIVE VARIABLE-BANDWIDTH ESTIMATOR [8].

density	n	best fix	Silver.	Botev	Chiu	Abram.	AD-bw	AD-Wien.
Gaussian	200	2.77	3.55	9.77	3.66	4.71	4.00	4.49
Gaussian	2000	0.52	0.62	1.53	0.58	0.84	0.60	0.56
Bimodal	200	4.19	4.63	10.64	5.19	4.34	5.55	5.46
Bimodal	2000	0.88	1.03	1.96	0.92	0.82	0.99	0.81
Kurtotic	200	23.32	65.59	56.06	24.96	25.68	25.24	25.31
Kurtotic	2000	4.31	20.48	9.56	4.57	3.30	4.55	3.79
Claw	200	23.60	48.53	41.84	42.20	49.05	32.42	29.56
Claw	2000	3.73	33.93	8.45	3.87	27.82	3.93	3.01
Asym. claw	200	15.38	22.79	21.74	18.42	22.52	18.33	19.35
Asym. claw	2000	3.88	13.10	5.51	4.24	11.38	4.16	4.54
Smooth comb	200	25.27	68.17	33.47	27.09	63.80	27.28	28.47
Smooth comb	2000	6.59	43.23	9.10	6.96	40.18	6.84	7.27
Discrete comb	200	26.52	92.98	33.93	28.47	88.83	29.75	29.04
Discrete comb	2000	5.06	50.33	10.22	8.35	42.20	5.16	4.43
Strongly skewed	200	25.14	91.71	59.25	28.74	66.33	29.47	31.51
Strongly skewed	2000	4.84	53.26	10.97	5.21	29.91	5.16	5.18

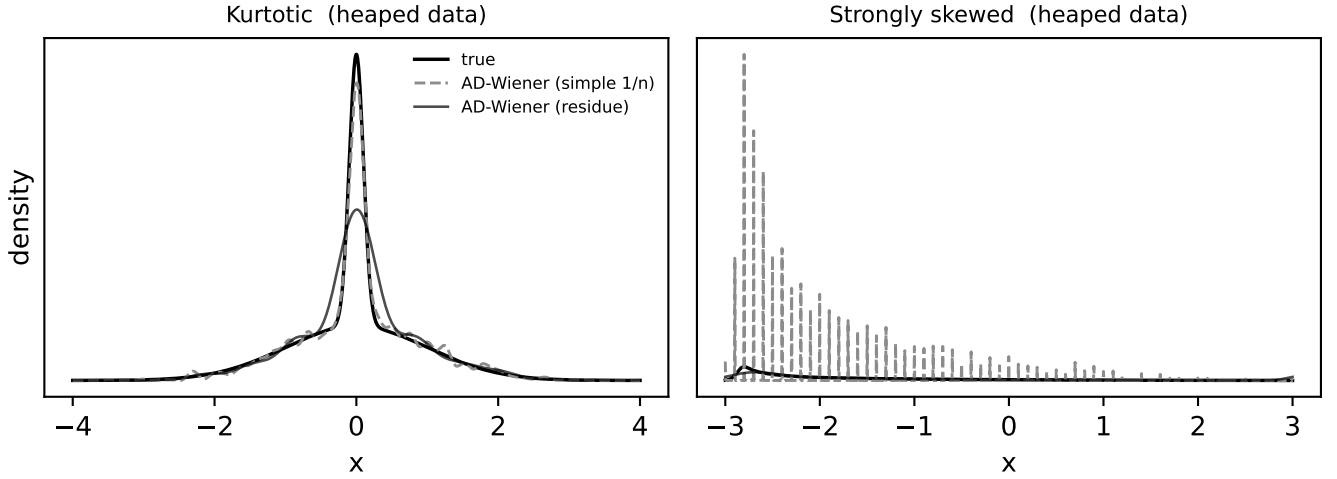


Fig. 1. Heaped data (rounded to 0.1). The simple $1/n$ Wiener estimate (dashed) throws spurious spikes at the rounding locations; the residue-floor Wiener estimate (solid) tracks the truth.

smooth, bimodal, and comb densities. A classical skewness or kurtosis statistic carries the same information and separates the asymmetric case from the symmetric spike.

The remedy follows from the matched-group viewpoint. A fixed grid makes the translation (cyclic) group the matched symmetry, which suits stationary, near-symmetric targets; a strongly asymmetric density violates that stationarity. Restoring it with a monotone symmetrizing transform [12], estimating with the AD-Wiener filter in the transformed domain, and mapping back through the Jacobian recovers the match. On the strongly skewed density this lowers the mean integrated squared error from 5.31 to 3.88 ($\times 10^3$), about a twenty-seven percent reduction, consistent across sample sizes and seeds. The transform is neutral on the symmetric kurtotic spike, which a monotone warp cannot symmetrize, and it distorts the regular structure of the comb densities, so it is applied only when the skewness probe fires; the comb densities, whose

coherent effective dimension is low, do not trigger it. This is the robustness the residue treatment was built for, applied here to the geometry of the sample rather than to the noise floor.

The construction is not confined to the spectral paradigm. The translation group is only one choice of matched symmetry, and the same group-averaging can be carried out under any group the data respect. Under the dilation (scaling) group, for instance, the eigenbasis is a wavelet multiresolution rather than the Fourier basis, and the coherent-versus-residue split of Section IV-A becomes wavelet-coefficient thresholding [17], [18]. We tested a translation-invariant realization of this dilation-group estimator and found that on the smooth Gaussian-mixture densities considered here it does not improve on the spectral construction, its advantage being confined to densities with genuine discontinuities; the spectral construction is therefore retained, with the matched-group viewpoint left as the organizing principle.

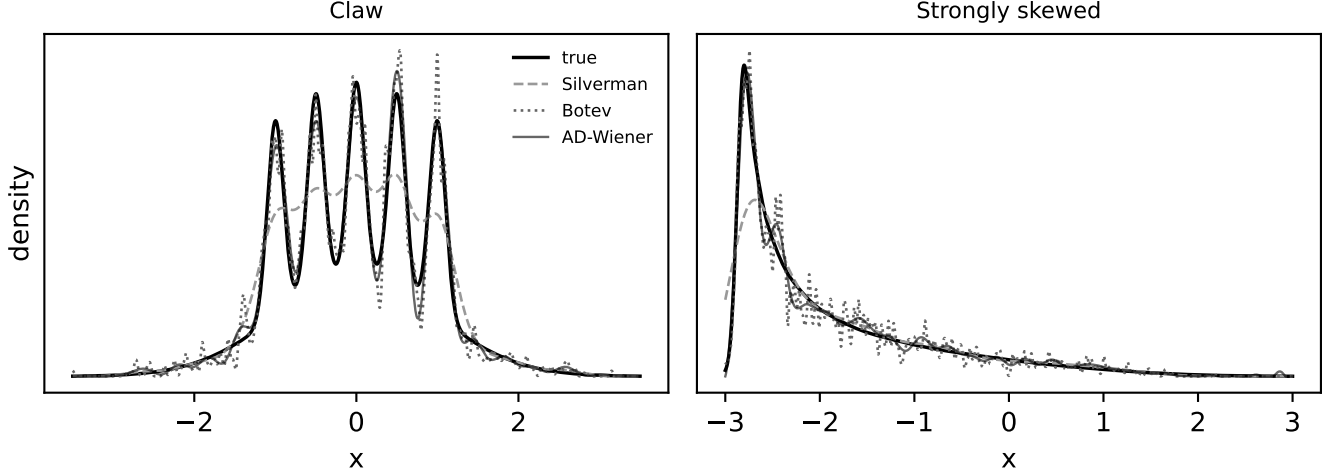


Fig. 2. Estimates at $n = 2000$. The adaptive AD-Wiener estimate resolves the structure that the rule of thumb smooths away, without the oscillation of the Sheather-Jones estimate.

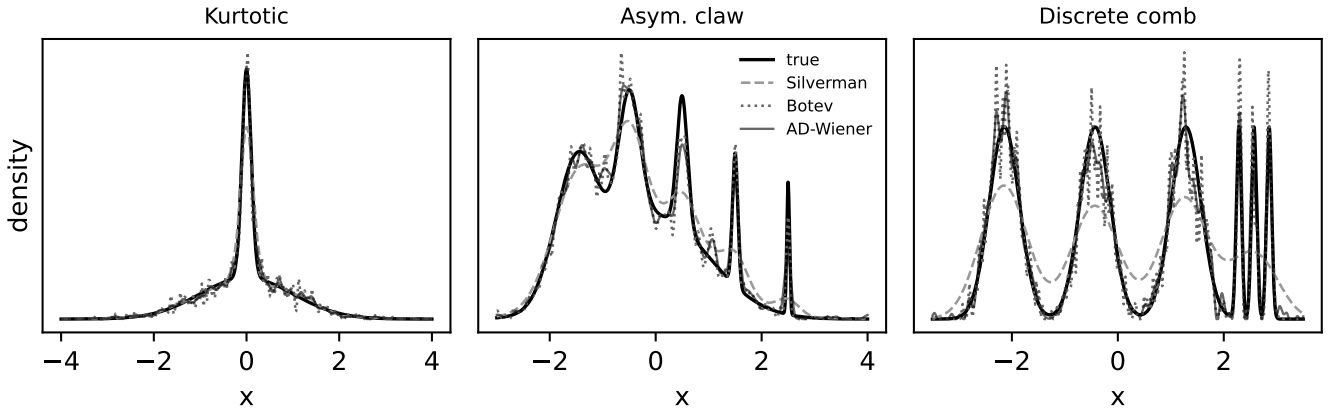


Fig. 3. Extreme multi-scale densities at $n = 2000$: a sharp spike over a broad base, peaks of decreasing width, and wide together with razor-thin peaks. Fixed-bandwidth methods cannot serve all scales at once; the per-frequency AD-Wiener filter can.

IX. DECONVOLUTION UNDER KNOWN MEASUREMENT ERROR

Suppose the observations are corrupted by measurement error, $Y = X + \varepsilon$, with ε of known distribution independent of X ; the target is the density of X , not of Y . The observed density is the convolution $f_Y = f_X * f_\varepsilon$, so an ordinary kernel estimator faithfully recovers the blurred f_Y and is biased for f_X . In the characteristic-function domain the convolution is a product, $\varphi_Y = \varphi_X \varphi_\varepsilon$, so deconvolution is division by the known noise characteristic function, carried out in the same domain in which the bandwidth selector and the AD-Wiener filter already operate [13], [19]. This is the spectral form of the Wiener filtering tradition the adaptive estimator descends from.

Division by φ_ε amplifies the floor. The flat $1/n$ sampling floor on the empirical characteristic function becomes the

frequency-shaped floor $(1/n)/|\varphi_\varepsilon(t)|^2$, which blows up where φ_ε decays, and this is the whole difficulty of deconvolution: the floor grows fastest exactly where one most wants to divide. The AD-Wiener taper carries over unchanged except that it is now applied against this amplified floor; the cutoff is the frequency at which the deconvolved power descends to meet the floor, found by comparing the empirical residual to the computed one and stopping where they agree. No bandwidth is set by hand. The error here is Laplace, ordinary smooth with $\varphi_\varepsilon(t) = 1/(1 + b^2 t^2)$; the harder super-smooth case such as Gaussian error, whose characteristic function decays exponentially and forces logarithmic rates [19], and the blind case in which φ_ε must itself be estimated, are left to future work.

The true density is bimodal so that the blur merges the two modes that deconvolution then recovers (Fig. 8, left): the naive estimator, seeing f_Y , smears the modes into a single

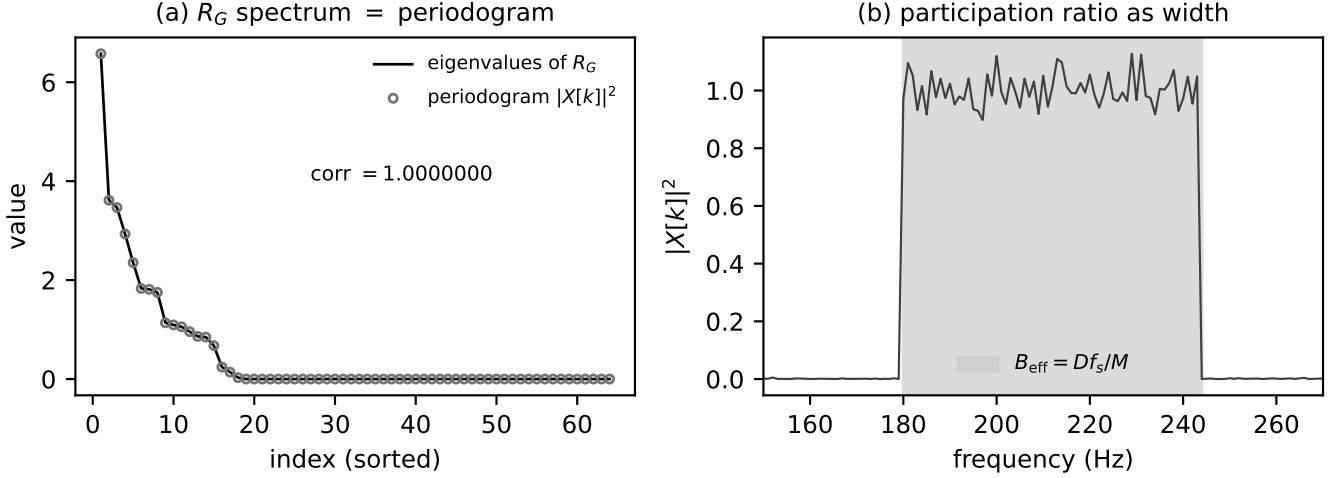


Fig. 4. The eigenvalues of the group-averaged covariance equal the periodogram; the participation ratio is the effective occupied width.

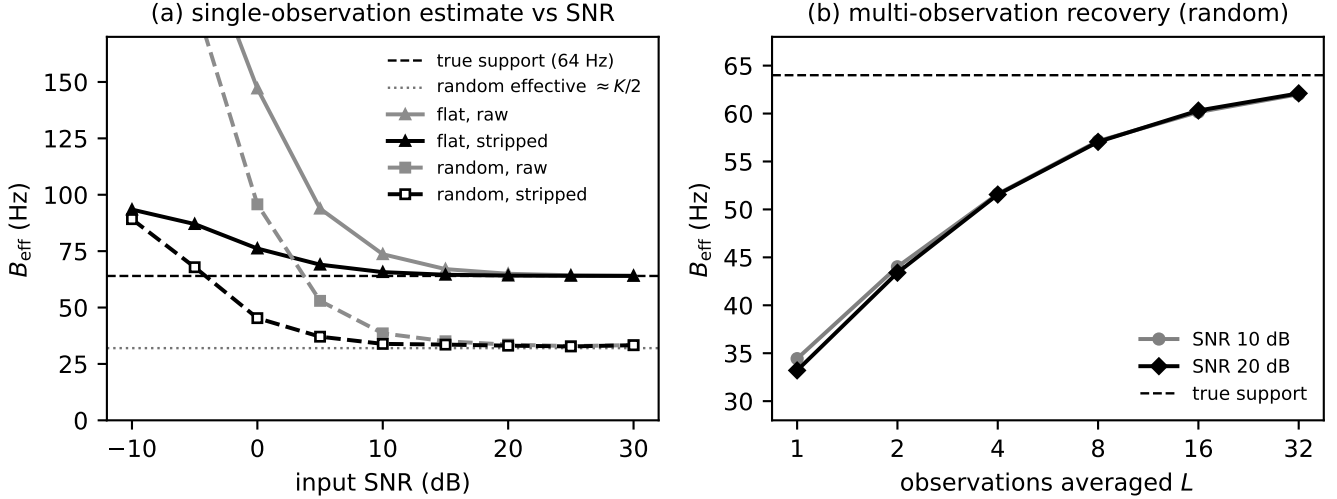


Fig. 5. Single-observation bandwidth versus signal-to-noise ratio, and recovery of the support bandwidth by averaging.

bump, while the deconvolution recovers them. Table III reports integrated squared error against the true f_X for the naive estimator, a standard deconvolving-kernel estimator at its oracle bandwidth, and the AD deconvolution. The naive estimator barely improves with sample size, since it is converging to the wrong, blurred density. The AD deconvolution is tuning-free and improves steadily, trailing the oracle-bandwidth estimator at the smallest samples, matching it by the moderate ones, and surpassing it at the largest, where its optimal taper outperforms a fixed kernel. The diagnostic that drives it is shown in Fig. 8, right: the deconvolved power meets the computed floor at t^* , and the stop fires there.

X. A LOCALLY PARTITIONED MIXED-MODE ESTIMATOR

The benchmark and the hard-density study show a division of competence: the AD-Wiener filter resolves sharp, highly

structured features such as claws and combs, while a Gaussian mixture is the more economical estimator on smooth densities. A density that places both kinds of structure in different regions of the support is served well by neither paradigm alone. A locally partitioned estimator removes the global commitment: it splits the support, applies each paradigm where that paradigm is strong, and joins the pieces. The join is the crux. A hard split is discontinuous at the boundary, which both distorts the estimate there and breaks the differentiability that gradient-based use needs. We join instead by a smooth partition of unity

$$w(x) = \frac{1}{2} \left(1 - \tanh \frac{x - x_0}{\delta} \right), \quad (4)$$

an elementary sigmoidal transition, a rescaled logistic function, that passes from one well to the left of the boundary x_0

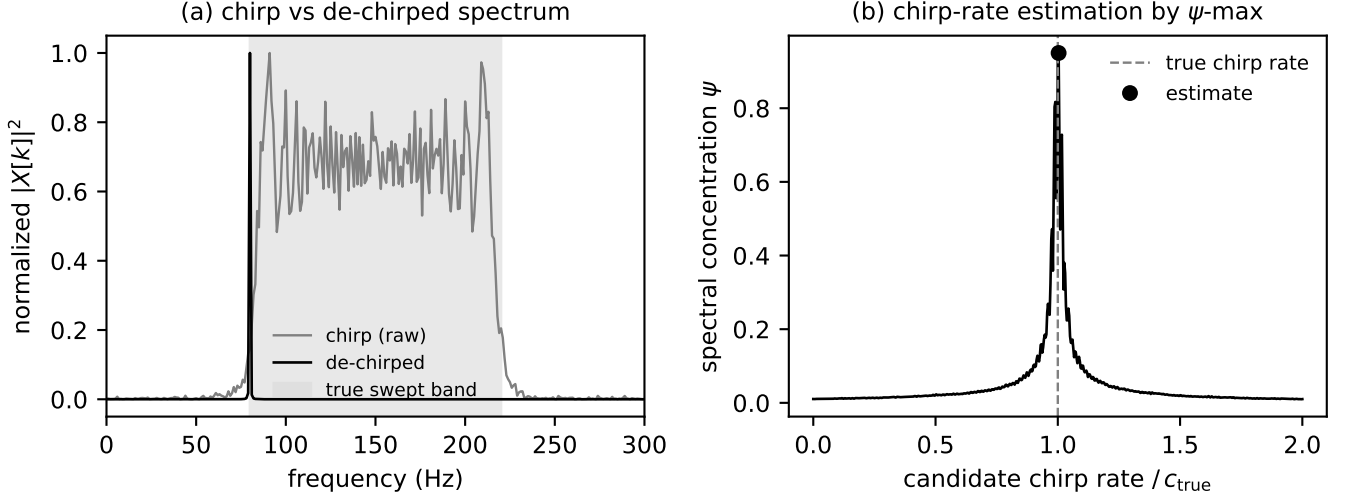


Fig. 6. For a chirp the cyclic estimate measures the swept bandwidth; the metaplectic chirp rate, recovered by spectral-concentration maximization, de-chirps to the instantaneous bandwidth.

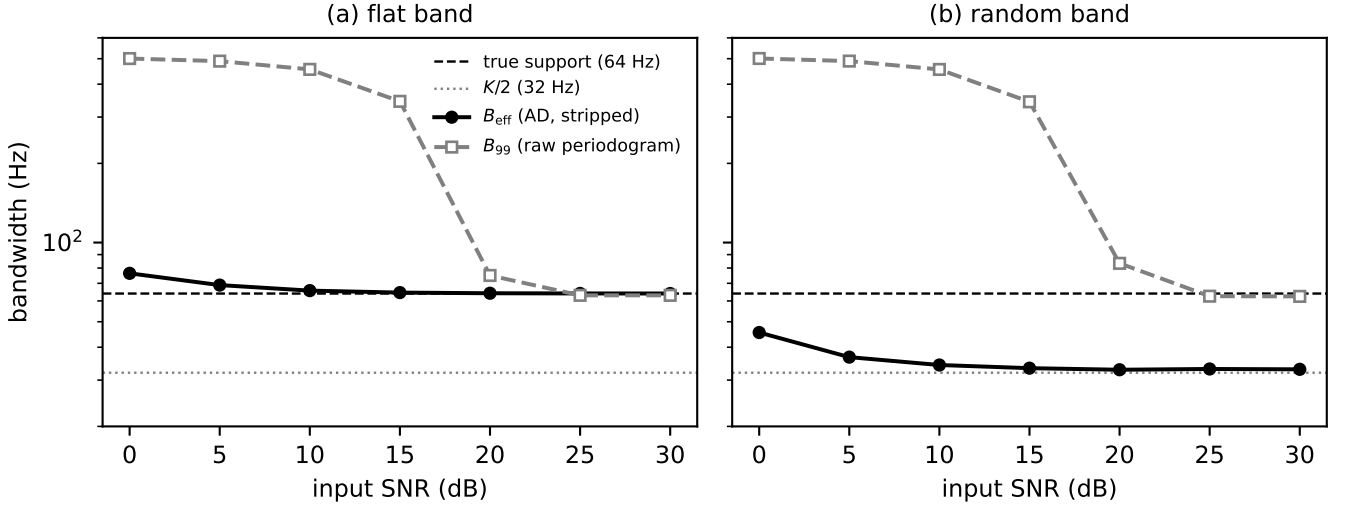


Fig. 7. The algebraic effective bandwidth against the conventional ninety-nine percent occupied bandwidth: the former measures an effective width and is far more robust to noise.

to zero well to its right across a width set by δ . The form is not drawn from a particular source; it is the standard smooth-step construction, and Lemma 1 records the two properties the join needs. The combined estimate $\hat{f} = w \hat{f}_{\text{left}} + (1-w) \hat{f}_{\text{right}}$ inherits the smoothness of its pieces and is continuous and differentiable across the boundary. The boundary location x_0 is supplied by the user; automatic detection of the seam from the data is left open.

Lemma 1 (Smooth, artifact-free join). *Let w be given by (4) with $\delta > 0$, and let $\hat{f}_{\text{left}}, \hat{f}_{\text{right}}$ be density estimates that are C^m on a neighborhood of x_0 . Then (i) $w \in C^\infty(\mathbb{R})$ with $0 < w(x) < 1$, so the blend $w \hat{f}_{\text{left}} + (1-w) \hat{f}_{\text{right}}$ is C^m there and C^∞ when the pieces are; and (ii) because $w(x) \in [0, 1]$*

pointwise, the blend is a pointwise convex combination of the two estimates, so $\min(\hat{f}_{\text{left}}, \hat{f}_{\text{right}}) \leq w \hat{f}_{\text{left}} + (1-w) \hat{f}_{\text{right}} \leq \max(\hat{f}_{\text{left}}, \hat{f}_{\text{right}})$ at every x . The join thus introduces no value above either estimate and none below both: no spurious spike and no dropout, for any boundary location and any width.

Proof. $\tanh \in C^\infty$ and $x \mapsto (x - x_0)/\delta$ is affine, so $w \in C^\infty$ with range $(0, 1)$; finite sums and products of C^m functions are C^m , which gives (i). For $a \in [0, 1]$, $au + (1-a)v$ lies between $\min(u, v)$ and $\max(u, v)$, which gives (ii). $\square$

The two estimators meet the hypothesis: the AD-Wiener estimate is band-limited, a trigonometric polynomial and so C^∞ except at the isolated points where the non-negativity

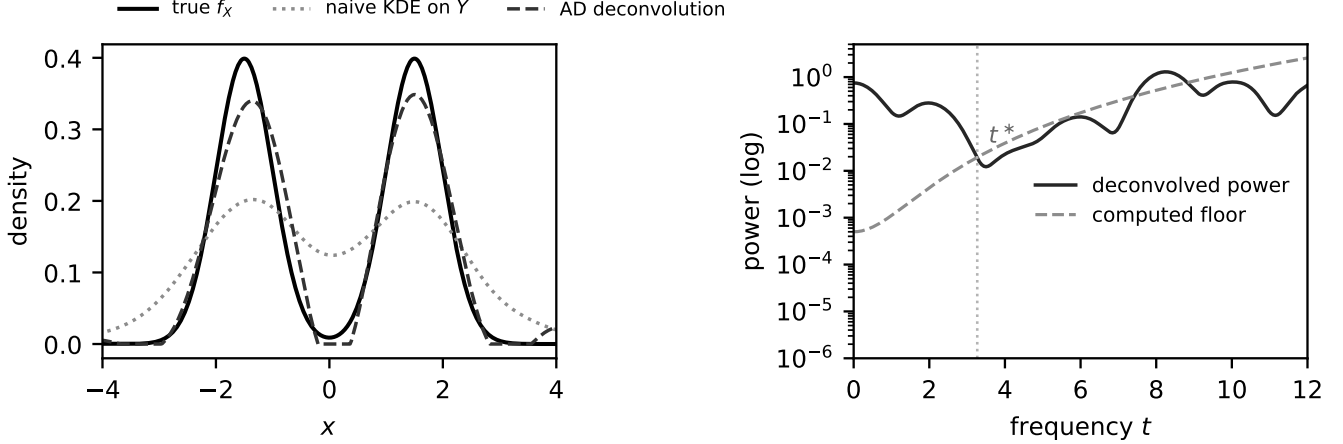


Fig. 8. Deconvolution under known Laplace error. Left: the naive estimator merges the two modes of f_X , while the AD deconvolution recovers them. Right: the deconvolved power meets the computed floor $(1/n)/|\varphi_\varepsilon|^2$ at t^* , where the stop fires; beyond it the rising power is amplified noise and is discarded.

TABLE II

MEAN INTEGRATED SQUARED ERROR ($\times 10^3$) ON HEAPED DATA (SAMPLES ROUNDED TO 0.1, $n = 2000$). THE FIXED $1/n$ FLOOR FAILS AS ALIASING LIFTS THE TRUE FLOOR; THE RESIDUE FLOOR ADAPTS. THE SIMPLE-STRIP WIENER ESTIMATE IS DESTROYED BY THE ROUNDING SPECTRUM, THE RESIDUE STRIP IS NOT.

density	AD-bw		AD-Wiener	
	simple	residue	simple	residue
Gaussian	0.70	0.60	0.65	0.44
Bimodal	1.01	0.93	0.88	0.73
Kurtotic	151.7	91.0	732.4	87.7
Claw	35.1	43.2	376.0	45.6
Asym. claw	85.1	13.7	2452.9	14.1
Smooth comb	39.6	20.6	3874.1	22.5
Discrete comb	23.1	19.7	2088.6	20.5
Strongly skewed	257.7	52.3	4712.3	62.5

TABLE III

DECONVOLUTION UNDER KNOWN LAPLACE ERROR ($b = 0.7$): INTEGRATED SQUARED ERROR ($\times 10^3$) AGAINST THE TRUE DENSITY f_X , MEAN OVER 20 REPLICATIONS. THE DECONVOLUTING-KERNEL BASELINE USES ITS ORACLE BANDWIDTH; AD DECONVOLUTION USES NONE.

n	naive KDE on Y	deconv. kernel (oracle)	AD deconvolution
250	89.1	32.0	49.3
500	81.6	23.0	38.1
1000	76.6	16.1	28.6
2000	70.8	14.2	20.0
4000	67.4	12.0	9.2

clip meets zero, and the Gaussian mixture is a finite sum of Gaussians and so C^∞ everywhere. The mixed-mode estimate is therefore differentiable to all orders on each open region, which is what gradient-based use requires; a single lemma states this completely, and no separate higher-order result is needed because the C^∞ conclusion already covers every order. Part (ii) was checked numerically by placing the boundary at adversarial locations, on a claw spike and inside a smooth peak among them, across widths from 0.05 to 1.5: the largest

departure of the blend from the envelope of the two estimates was at the level of machine precision.

To test the construction we take a combined target, a broad Gaussian-mixture base with a claw of narrow spikes superimposed near the origin, split at its seam $x_0 = 0$, and evaluate all four assignments of the two paradigms to the two halves. Fidelity to the specified, ideal density is measured two ways: the Kullback-Leibler divergence [20] of the ideal input from the estimated output, and the symmetric Jensen-Shannon divergence [21], both with lower values better. Table IV and Figure 9 give the outcome. Assigning the mixture to the smooth half and AD-Wiener to the claw half is best by both divergences, and it improves on the best global choice, AD-Wiener on the whole support, by about a third in Kullback-Leibler divergence (0.007 against 0.011): placing the mixture where it is exact removes the small ripple AD-Wiener leaves on the smooth half. The reversed assignment, AD-Wiener on the smooth half and the mixture on the claw, is the worst of the four, worse even than the mixture everywhere, because it places each paradigm where it is weakest. The ordering confirms that a correct local partition improves on every single global estimator, and that the cost of a wrong partition is real. Read as a check on the global default, the same table supports AD-Wiener as the global default: among the estimators applied uniformly, AD-Wiener has the lowest divergence by a wide margin, 0.011 in Kullback-Leibler against 0.064 for the mixture, and only a correct local partition lowers it further.

The smooth join leaves no discontinuity at the seam, visible in Figure 9. What the four-combination experiment fixes by hand is the partition itself. The boundary can be supplied manually, one interval and one method at a time; in the general case, where the target density is unknown, choosing the boundary from the data alone is a model-selection problem this paper leaves open. When the target is known, however, as in the synthetic-data quality-checking setting of Section XII, the

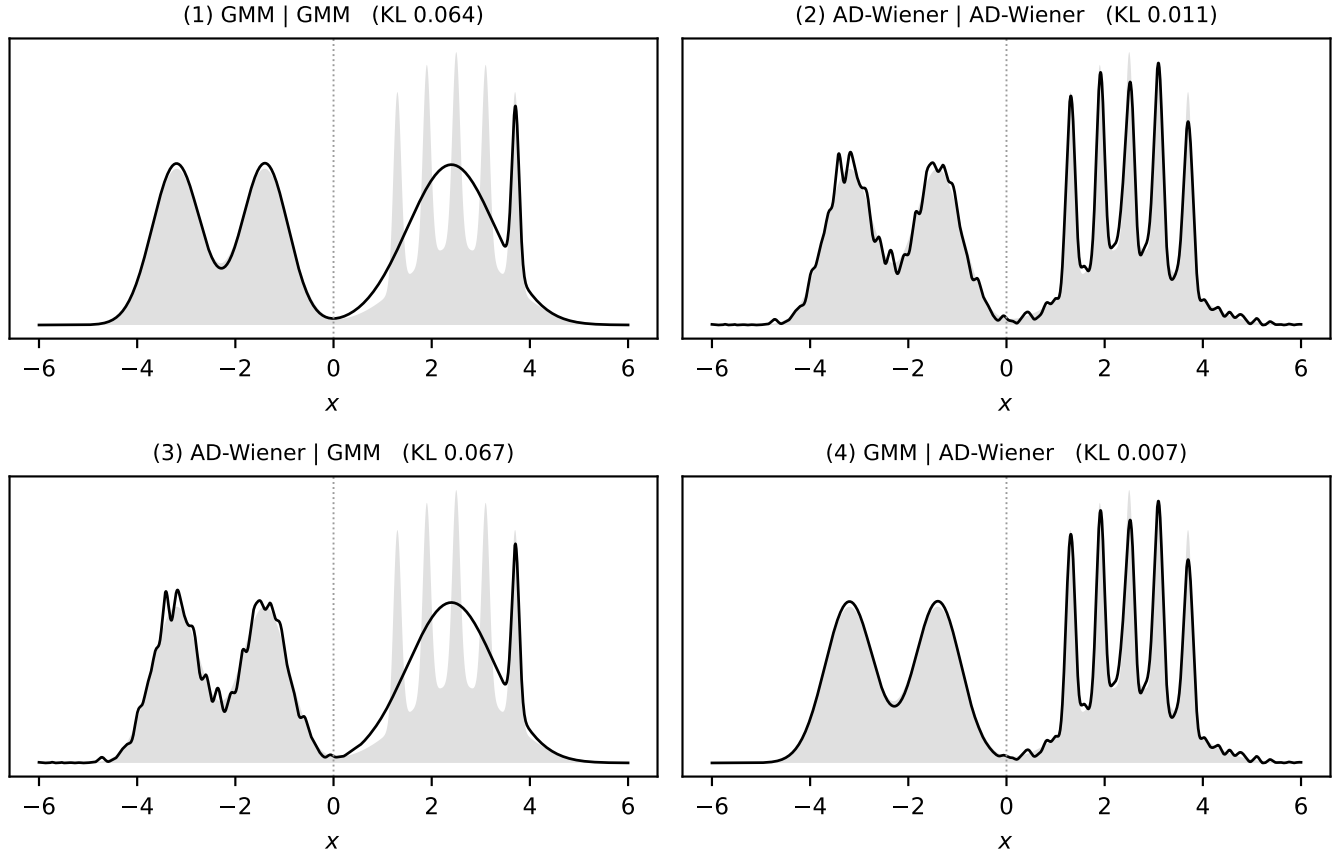


Fig. 9. The four method assignments on the combined smooth-mixture and claw target, joined by a smooth partition of unity at the seam $x_0 = 0$ (dotted line); the gray fill is the ideal target and the parenthetical value is the Kullback-Leibler divergence. (1) the mixture misses the claw; (2) AD-Wiener resolves the claw and roughly tracks the smooth half; (3) the reversed assignment is worst, each paradigm placed where it is weakest; (4) the mixture on the smooth half and AD-Wiener on the claw is best, capturing both. The join carries no visible discontinuity at the seam.

TABLE IV

FIDELITY OF THE FOUR METHOD ASSIGNMENTS ON THE COMBINED SMOOTH-MIXTURE AND CLAW TARGET ($N = 8000$, EXACT GENERATION, PARTITION AT THE SEAM $x_0 = 0$): KULLBACK-LEIBLER DIVERGENCE OF THE IDEAL INPUT FROM THE ESTIMATED OUTPUT, AND THE SYMMETRIC JENSEN-SHANNON DIVERGENCE, BOTH LOWER BETTER. THE MIXTURE ON THE SMOOTH HALF WITH AD-WIENER ON THE CLAW HALF IS BEST AND IMPROVES ON EVERY GLOBAL CHOICE; THE REVERSED ASSIGNMENT IS WORST.

smooth half claw half	$KL(f \parallel \hat{f})$	$JS(f, \hat{f})$
GMM GMM	0.0643	0.0222
AD-Wiener AD-Wiener	0.0105	0.0025
AD-Wiener GMM	0.0666	0.0229
GMM AD-Wiener	0.0074	0.0015

partition can be chosen automatically by a rule that needs no tuning: in each bin of the support retain whichever estimator has the lower local Kullback-Leibler divergence to the target, merge adjacent bins of like choice into segments, and join the segments by (4) at the boundaries where the choice changes. This places the boundaries and assigns the methods from the specified target alone.

Table V exercises the automatic rule across five generated

targets with known densities, from a smooth Gaussian mixture through a five-spike claw to a density that alternates smooth bumps with claw combs. On the smooth and the kurtotic targets the automatic estimator recovers the mixture and matches it; on the claw, the half-and-half, and the alternating targets it lowers the Kullback-Leibler divergence of the AD-Wiener default by factors of two to four, by placing the mixture on the smooth stretches and AD-Wiener on the sharp ones, and it is never worse than the default in that divergence. The rule selects by Kullback-Leibler divergence, so that divergence falls in every case; the Jensen-Shannon divergence, which the rule does not target, also falls except on the alternating density, where it is marginally above the default. The alternating target is the clearest case. Figure 10 shows AD-Wiener tracking the combs but rippling on the smooth bumps, while the automatic mixed-mode, with its detected boundaries marked, gives clean bumps and sharp combs together.

The rule above uses the specified target, which the generator and the beacon stream of Section XVIII both provide, and there it is automatic and well behaved. Outside that setting the target is unknown, and a partition read from the data alone runs into a circularity: detecting where a sharp feature begins and ends is the boundary-placement problem restated. A purely

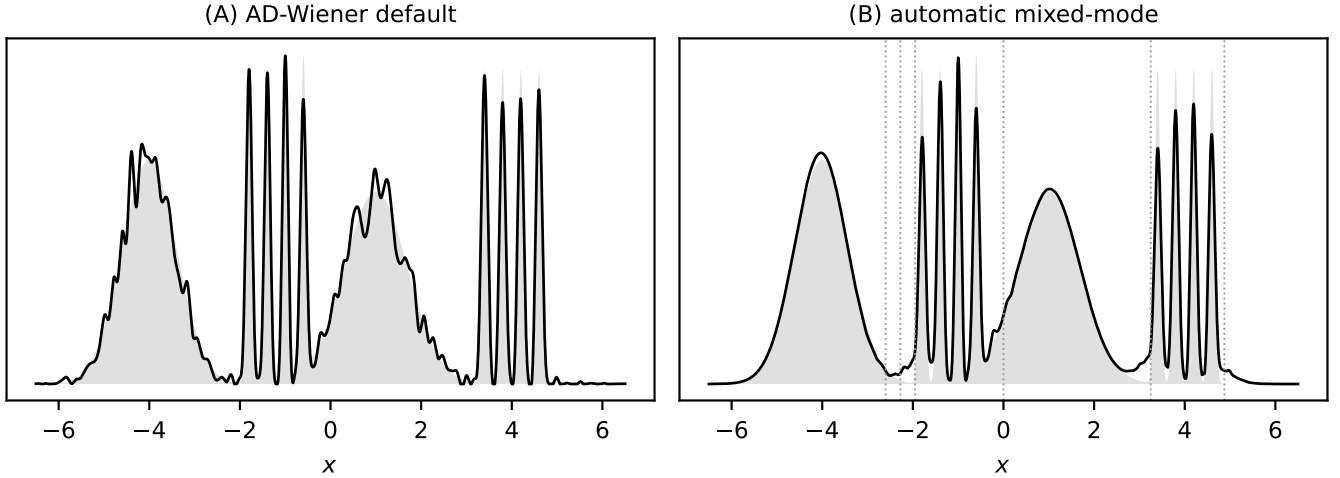


Fig. 10. The alternating target of smooth bumps and claw combs. (A) the AD-Wiener default resolves the combs but ripples on the smooth bumps; (B) the divergence-guided automatic mixed-mode, with its detected boundaries dotted, applies the mixture on the smooth bumps and AD-Wiener on the combs, giving clean bumps and sharp combs at once. The gray fill is the specified target.

TABLE V

AD-KDE ACROSS FIVE GENERATED TARGETS ($N = 8000$, MEAN OVER FIVE SEEDS): KULLBACK-LEIBLER AND JENSEN-SHANNON DIVERGENCE OF THE SPECIFIED TARGET FROM THE ESTIMATE, FOR THE MIXTURE, THE AD-WIENER DEFAULT, AND THE DIVERGENCE-GUIDED AUTOMATIC MIXED-MODE. THE AUTOMATIC ESTIMATOR MATCHES THE MIXTURE ON THE SMOOTH TARGETS AND LOWERS THE AD-WIENER DEFAULT ON THE HETEROGENEOUS ONES; THE LOWEST VALUE IN EACH ROW AND DIVERGENCE IS IN BOLD.

target	GMM		AD-Wiener		mixed-auto	
	KL	JS	KL	JS	KL	JS
bimodal	0.0003	0.0001	0.0047	0.0008	0.0003	0.0001
kurtotic	0.0002	0.0001	0.0113	0.0015	0.0002	0.0001
claw	0.0442	0.0160	0.0090	0.0013	0.0022	0.0008
half-and-half	0.0728	0.0255	0.0115	0.0021	0.0050	0.0018
alternating	0.2078	0.0797	0.0293	0.0040	0.0218	0.0093

data-driven surrogate, a cross-validated local-likelihood comparison of the two estimators, bears this out. It is unreliable in the low-density valleys between sharp features, where the held-out comparison is noisy and intermittently prefers the mixture, removing exactly the structure AD-Wiener is needed for; and it cannot separate a smooth base from the closely-spaced spikes that ride on it, the claw, because the base and the spikes occupy the same support. The framework therefore takes two routes that avoid the circularity rather than confront it. When the target is known the divergence-guided rule applies directly. When it is not, the boundaries are supplied by the user, who often knows where the regimes change and may legitimately want a sharp region resolved by AD-Wiener in one place and smoothed by the mixture in another, with the method inside each user-given region selected automatically by held-out likelihood. The fully automatic default, which needs neither a target nor a boundary, is the superposition of Section XI: rather than partition the support it decomposes the density into a smooth base and a sharp residual, and so

dissolves the boundary question entirely.

XI. SUPERPOSITION OF A SMOOTH BASE AND A SHARP RESIDUAL

The mixed-mode estimator of Section X assigns one method to each region of the support, which presumes the two regimes are spatially separated: smooth here, sharp there. A claw of narrow spikes sitting on a broad Gaussian base violates that presumption. The base and the spikes occupy the same locations, so no spatial boundary separates them; a partition either places the spikes in a region it has called smooth, where the mixture erases them, or it splits the spike cluster and fits the pieces apart, as a midpoint split of the support does whenever the cut falls inside the cluster. The structure is superimposed, not adjacent, and calls for a decomposition rather than a partition.

Write the density as a sum,

$$\hat{f}(x) = \hat{b}(x) + \hat{s}(x), \quad (5)$$

a smooth base $\hat{b}$ and a sharp residual $\hat{s}$, each fit by the method suited to it and added back. A Gaussian mixture is fit by the order selection of Section X and its components are split by width at the smoothness scale $\theta = ch$, with h the Silverman bandwidth: the components of standard deviation at least θ form the base $\hat{b}$, the narrower components are set aside. The base then omits whatever is sharper than θ , so the residual mass $p - \hat{b}$ carries that sharp structure, and the AD-Wiener filter of Section V applied to the residual recovers it as $\hat{s}$, keeping the coherent part above the $1/n$ floor and discarding the rest. No region is chosen and no boundary is detected: the same two estimators act on the whole support, one on the smooth part and one on what it leaves behind. Figure 11 shows the decomposition on the claw, the mixture base, the band-limited residual carrying the five spikes, and their sum against the target.

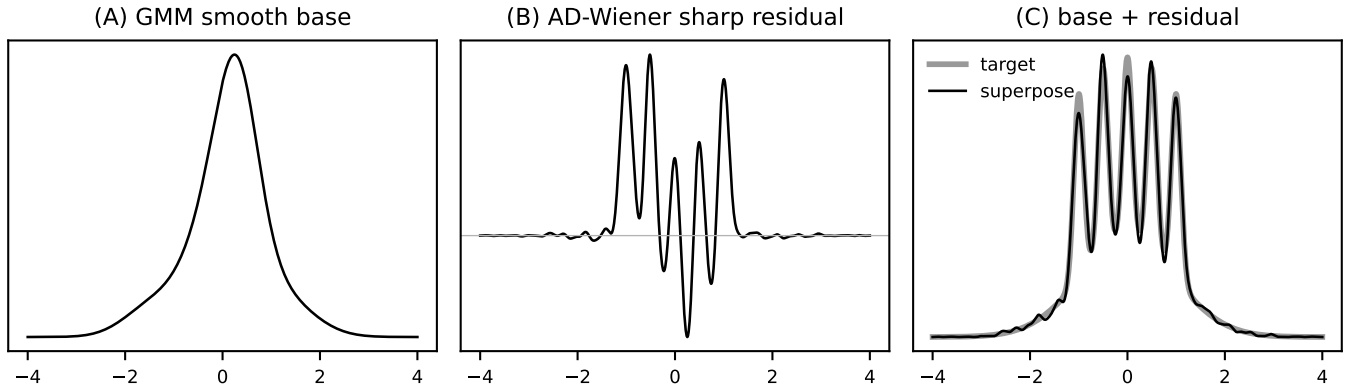


Fig. 11. Superposition on the claw. (A) The Gaussian-mixture base captures the broad envelope. (B) The AD-Wiener filter on the residual mass recovers the five spikes the base omits, with small negative lobes where the base slightly over-predicts. (C) Their sum recovers the target without any spatial boundary.

TABLE VI
SUPERPOSITION AGAINST ITS TWO INGREDIENTS: KULLBACK-LEIBLER DIVERGENCE TO THE TARGET, MEAN OVER FIVE SEEDS AT $N = 8000$, SMOOTHNESS SCALE $c = 1.5$. SUPERPOSITION TRACKS THE BETTER INGREDIENT ON EVERY TARGET.

target	GMM	AD-Wiener	superposition
bimodal	0.0003	0.0047	0.0003
kurtotic	0.0002	0.0113	0.0116
claw	0.0442	0.0090	0.0091
half-and-half	0.0728	0.0115	0.0117
alternating	0.2078	0.0293	0.0293

Table VI reports the result. Superposition tracks the better of its two ingredients on every target without choosing between them. On the smooth bimodal and kurtotic densities it matches the mixture, where AD-Wiener alone carries ripple; on the spiked claw, the half-and-half density, and the alternating comb it matches AD-Wiener, where the mixture alone cannot place the spikes. It does not merely avoid the worse method: it recovers both strengths at once, a single estimator that is mixture-accurate on the smooth part and band-limited-accurate on the sharp part, which neither ingredient is alone. The contrast with the spatial mixed-mode is sharpest on the claw, the case where a midpoint split is forced to break the spike cluster. There superposition matches AD-Wiener, 0.0091 against 0.0090, rather than degrading it, precisely because it never tries to separate the base from the spikes in space. Two limits are worth stating plainly. Superposition ties AD-Wiener on the pure-spike cases rather than beating it, since the band-limited residual carries the ripple AD-Wiener would; and on a single isolated sharp Gaussian, the kurtotic spike, routing that one component through the residual gives band-limited rather than mixture-exact accuracy, 0.0116 against the mixture’s 0.0002. It does not reach the divergence-guided rule of Section X, which is given the target, but unlike that rule it needs neither a target nor a boundary, and unlike any spatial partition it does not degrade on superimposed structure.

The real-data sections bear this out. Where the sharp structure is well sampled and a single estimator already resolves it,

superposition matches the better estimator without improving on it: it ties AD-Wiener on the CMS dimuon resonances of Section XVI (held-out negative log-likelihood 0.314) and on the SDSS redshift features of Section XVII (−2.009), and on the leptokurtic CRSP returns of Section XIV its residual finds no structure above the floor, so it reduces to the mixture and reproduces the mixture’s tail-risk accuracy. Its one real-data gain is on heaped data, Section XV: there the residual filter reads the rounding comb as an artifact below the coherence floor and discards it, so the smooth base recovers the de-heaped density at a nearly constant error as the heaping coarsens, overtaking even the residue floor at the coarsest grids. The pattern is consistent: superposition equals the best available estimator on every real dataset and improves on all of them only where robustness to an artifact is what the data demand.

XII. A SYNTHETIC-DATA APPLICATION: DENSITY FIDELITY UNDER CONTROLLED GENERATION ERROR

A common task in applied machine learning is to generate a dataset that follows a specified nonparametric density. A practitioner supplies a target density f and a sample count N , a generator emits N points meant to follow f , and the practitioner asks how closely the realized data match the request. Kernel density estimation answers that question by recovering the realized density from the generated sample and measuring its departure from f . This section frames that measurement as an application of the present estimators. It is a validation of the measurement procedure, not a ranking of generators: the generator used here is a parameterized reference whose departure from the target is known by construction, so the question is whether the recovered fidelity tracks the true departure, and which estimator recovers it with least bias.

The reference generator draws exactly from an arbitrary supplied density by inverse-cumulative sampling of its grid, then introduces a controlled defect by the Huber contamination model [22] $G_\varepsilon = (1 - \varepsilon)f + \varepsilon c$, where c is a contaminant such as a narrow spurious mode or a mis-modeled tail. With probability $1 - \varepsilon$ a draw is exact and with probability ε it

Algorithm 1 Reference generator with a known error knob

Require: target density f on grid x_g ; count N ; error $\varepsilon \in [0, 1]$; contaminant c ; jitter $\sigma_j \geq 0$

- 1: $G \leftarrow (1 - \varepsilon)f + \varepsilon c$ $\triangleright$ realized law; departs from f by $\varepsilon \text{TV}(f, c)$
- 2: $F \leftarrow \text{cumtrapz}(G, x_g)$; $F \leftarrow F/F_{\text{end}}$ $\triangleright$ normalized CDF
- 3: draw $u_1, \dots, u_N \sim \text{Unif}(0, 1)$
- 4: $d_i \leftarrow F^{-1}(u_i)$ by interpolation $\triangleright$ exact inverse-CDF draws
- 5: **if** $\sigma_j > 0$ **then**
- 6: $d_i \leftarrow d_i + \mathcal{N}(0, \sigma_j^2)$ $\triangleright$ optional smear
- 7: **end if**
- 8: **return** samples $d_1, \dots, d_N$ and the closed-form realized law G

comes from c , so the realized law departs from the target by $\varepsilon \text{TV}(f, c)$ in total variation and ε is the literal error knob; an optional jitter convolves the draws with a small Gaussian for a generator that smears rather than spikes. Algorithm 1 states the procedure. The sample count is bounded to $N \in [2^9, 2^{18}]$: below a few hundred points the comparison is sampling-noise limited, as the SDSS pencil beam of Section XVII shows, and above a few hundred thousand all estimators converge and the comparison is uninformative. Because G_ε is built rather than observed, its true departure from f is known in closed form and serves as ground truth, which the real-data sections of this paper cannot offer.

The procedure admits any target the practitioner supplies. One target is the combined smooth-and-sharp density that the mixed-mode and superposition estimators of Sections X and XI were built to handle. Its left half is a smooth low-order Gaussian mixture, the regime an adaptive Gaussian mixture recovers most accurately, and its right half is a Marron-Wand claw, a broad base carrying several narrow spikes that a bounded-order mixture cannot represent but the AD-Wiener estimator resolves. Generated exactly ($\varepsilon = 0$) and estimated by a single global estimator, this target produces the split of Table VII and Figure 13: the mixture is best on the smooth half by roughly an order of magnitude, AD-Wiener is best on the claw half by more than an order of magnitude, and neither is best on both. The current estimator selects between the two paradigms through a single global effective dimension, here $D_2 = 2.5$, so it must commit the whole support to one of them. The same global cutoff that resolves the claw also leaves a faint ripple on the smooth half, visible in panel (A), the price of a single decision for a spatially heterogeneous density.

This tie-case fixes the global default used throughout the paper. When no paradigm dominates globally, breaking the tie toward AD-Wiener bounds the worst case, since its full-support error here is 1.2×10^{-3} against the mixture's 20×10^{-3} , and a mixture chosen wrongly fails on any sharp or comb structure it cannot parameterize. AD-Wiener is therefore the default estimator throughout, with the mixture selected only when the effective dimension is clearly low, and the software accepts a manual override that forces either estimator

TABLE VII

INTEGRATED SQUARED ERROR ($\times 10^3$) ON THE `HALFHALL` TARGET ($N = 8000$, EXACT GENERATION), BY HALF. THE MIXTURE IS BEST ON THE SMOOTH HALF, AD-WIENER ON THE CLAW HALF; NEITHER IS BEST ON BOTH, WHILE THE GLOBAL EFFECTIVE DIMENSION $D_2 = 2.5$ SELECTS ONE PARADIGM FOR THE WHOLE SUPPORT.

estimator	smooth half	claw half	full
naive KDE	2.07	24.13	26.20
mixture (GMM)	0.09	20.14	20.23
AD-Wiener	0.43	0.75	1.18

on the whole support or applies a user-named method on each of a set of intervals, a partition the practitioner controls directly. The same heterogeneity is what the locally partitioned estimator of Section X and the superposition of Section XI resolve, by partitioning or decomposing the support rather than committing it to one paradigm.

The fidelity question proper is whether the recovered density's departure from the requested target tracks the generator's true departure. Sweeping the error knob ε and comparing the recovered total variation $\text{TV}(\hat{f}, f)$ with the known $\text{TV}(G_\varepsilon, f) = \varepsilon \text{TV}(c, f)$ gives Figure 12. On a mixture-representable target (kurtotic) every estimator recovers the departure, the mixture and AD-Wiener almost exactly and the ordinary estimate with a small positive bias from its own smoothing. On the multi-scale `halfhalf` target the estimators separate: AD-Wiener tracks the true departure across the whole range, while the ordinary and mixture estimates carry a large constant bias, roughly 0.12 to 0.18 in total variation, that does not vanish at $\varepsilon = 0$ because their model mismatch on the claw is itself read as a departure. For measuring how faithfully a generator reproduces a spiky or multi-scale target, AD-Wiener is thus the estimator that reports the true error rather than its own.

Differentiability matters for downstream use, since model builders and digital-twin constructions often optimize against an estimated density by gradient descent, and the spiky targets above are exactly where a non-differentiable estimate would hurt. Both estimators are differentiable. A Gaussian mixture is a finite sum of Gaussians and is infinitely differentiable everywhere, with closed-form gradients. The AD-Wiener estimate is band-limited, the Wiener taper setting all frequencies above the cutoff to zero, so its reconstruction is a trigonometric polynomial and is likewise infinitely differentiable; the only non-smoothness comes from the non-negativity clip, which produces isolated kinks at the finite set of points where the reconstruction crosses zero and is absent wherever the band-limited estimate stays non-negative. A manually partitioned estimate, by contrast, is in general discontinuous at the interval boundaries, which is the matching the automatic locally partitioned estimator must perform.

XIII. A FINANCIAL APPLICATION: TAIL RISK UNDER LEPTOKURTOSIS

Asset returns are sharply peaked and heavy-tailed, the leptokurtic shape the kurtotic density caricatures. We model daily returns as a calm and turbulent normal mixture, ninety

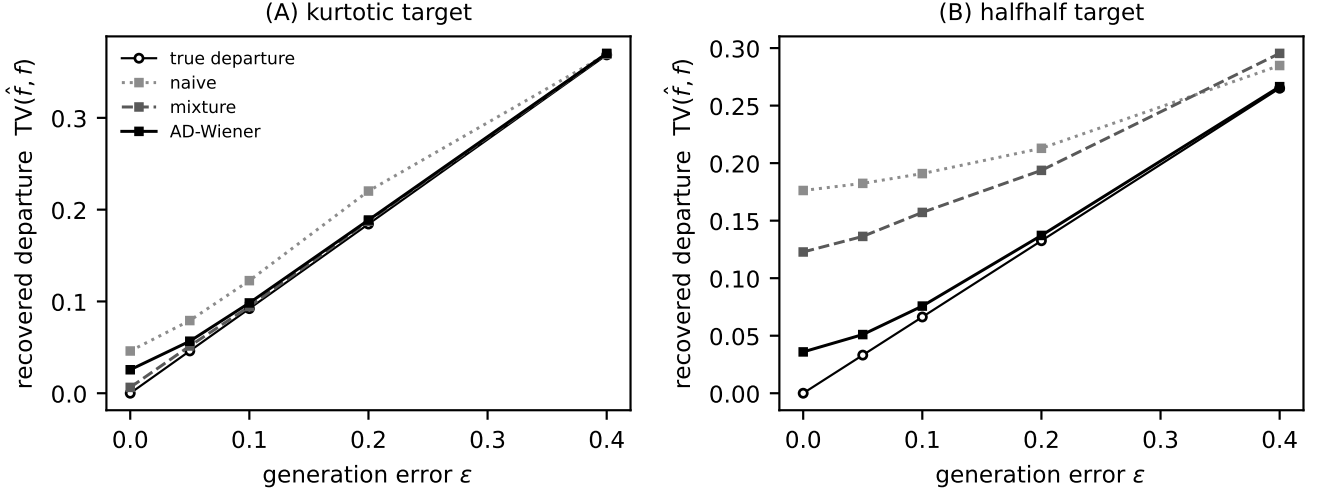


Fig. 12. Recovered departure $TV(\hat{f}, f)$ against the generation error ε , mean over five seeds at $N = 8000$; the open-circle line is the known true departure $TV(G_\varepsilon, f)$. (A) On the mixture-representable kurtotic target every estimator tracks the true departure, the ordinary estimate slightly high. (B) On the multi-scale halfhalf target AD-Wiener tracks the true departure while the ordinary and mixture estimates carry a constant model-mismatch bias that persists at $\varepsilon = 0$.

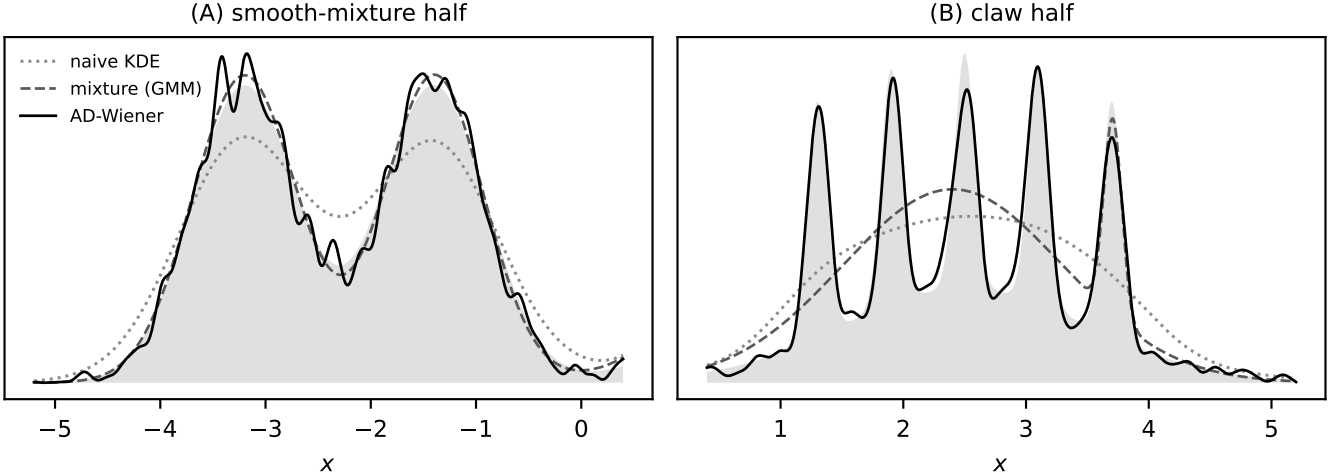


Fig. 13. The halfhalf target, generated exactly ($\varepsilon = 0$) and estimated globally. (A) On the smooth-mixture half the adaptive mixture tracks the target while AD-Wiener slightly over-resolves it and the ordinary estimate over-smooths. (B) On the claw half AD-Wiener resolves the narrow spikes that the bounded-order mixture and the ordinary estimate both miss. A single global estimator is optimal on neither half.

percent at a 0.8% daily volatility and ten percent at 3%, an excess kurtosis of 8.7, and estimate the return density and its left-tail risk, the Value-at-Risk and Expected Shortfall, from $n = 1000$ returns. A Gaussian fit is the standard parametric baseline; the spectral AD-Wiener estimator and an ordinary kernel estimate are nonparametric; the adaptive mixture is supplied through the same plugin interface.

The Gaussian fit cannot represent the peak and the tails at once and underestimates the one-percent Value-at-Risk by about a full percent of daily return and the Expected Shortfall by twice that (Table VIII, Fig. 14), the direction that matters for risk. The spectral AD-Wiener and ordinary kernel

estimators track the tail far better. The adaptive mixture, which matches the generating structure, is best on both the density and the tail risk, recovering the two regimes and cutting the density error more than fiftyfold against the Gaussian. The kurtotic case, read as a return distribution, is exactly where the algebraic and mixture estimators earn their value.

XIV. REAL-DATA VALIDATION ON CRSP RETURNS

The synthetic study of Section XIII is validated on real daily returns from the CRSP database over 2005–2024: the value- and equal-weighted market indices, the S&P 500, and a panel of ten large common stocks, 5,033 trading days each.

TABLE VIII

LEPTOKURTIC DAILY RETURNS (EXCESS KURTOSIS 8.7): INTEGRATED SQUARED ERROR OF THE DENSITY AND ABSOLUTE ERROR OF THE ONE- AND FIVE-PERCENT VALUE-AT-RISK AND ONE-PERCENT EXPECTED SHORTFALL, AGAINST THE TRUE VALUES. TAIL ERRORS IN BASIS POINTS OF DAILY RETURN. ISE IS MEAN $\pm$ STANDARD DEVIATION OVER 200 REPLICATIONS.

method	ISE ($\times 10^3$)	VaR _{1%}	VaR _{5%}	ES _{1%}
Gaussian	2341 $\pm$ 592	103	39	204
KDE	133 $\pm$ 74	1.3	3.5	4.3
AD-Wiener	250 $\pm$ 136	11.1	0.4	24.3
Mixture	41.5 $\pm$ 40.1	4.8	0.3	3.8

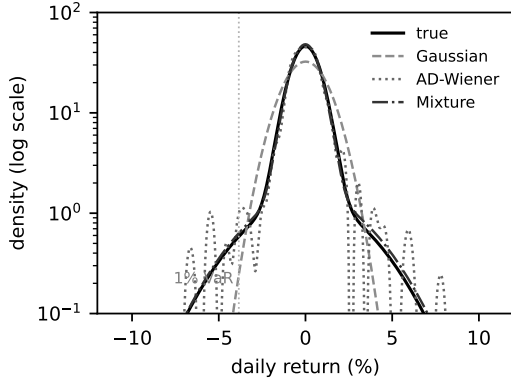


Fig. 14. Estimated return densities on a logarithmic scale. The Gaussian fit collapses the peak and underweights the tails; the mixture and AD-Wiener estimators track both.

Extraction and analysis are scripted, so the study reruns on any CRSP subscription: an extraction step pulls the daily series into a `data/` directory, and an analysis step applies the four estimators and writes the results as JSON into a `results/` directory. Because the true law is unknown for real returns, each estimator is measured against the historical (empirical) quantile and shortfall.

A. Asset return distributions

Every series is sharply peaked and heavy-tailed. Fig. 15 sets the S&P 500 daily return density against its histogram: the Gaussian fit collapses the peak and underweights the tails, while the AD-Wiener and mixture estimators track both, exactly as on the synthetic kurtotic density.

B. Risk estimation

Table IX reports annualized volatility, skewness, excess kurtosis, and a Hill left-tail index per series. Excess kurtosis runs from 5.4 to 19.7 and the Hill index from 2.5 to 3.5, tails far heavier than Gaussian across the panel, with the market indices mildly left-skewed and the heaviest tail in the bank stock.

C. Value-at-Risk and tail risk

Table X aggregates accuracy as the mean absolute deviation from the historical reference across the thirteen series. The

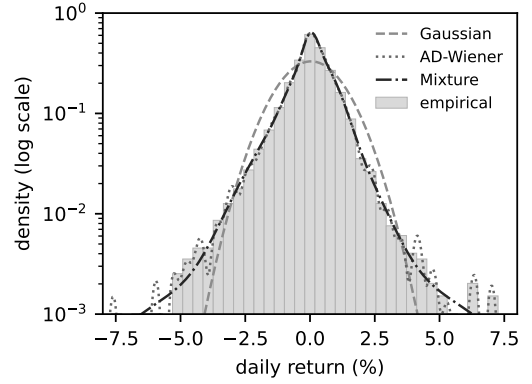


Fig. 15. S&P 500 daily return density (log scale) against the empirical histogram. The Gaussian fit misses the peak and the tails; the AD-Wiener and mixture estimators track both.

TABLE IX

RISK SUMMARY FOR THE CRSP SERIES, DAILY RETURNS 2005–2024 ($n = 5,033$ EACH): ANNUALIZED VOLATILITY, SKEWNESS, EXCESS KURTOSIS, AND HILL LEFT-TAIL INDEX.

series	vol (ann.)	skew	exc. kurt.	Hill
market (vw)	0.191	−0.38	11.9	2.77
market (ew)	0.181	−0.56	11.1	2.59
S&P 500	0.192	−0.26	12.8	2.70
AAPL	0.321	−0.02	5.4	3.45
MSFT	0.271	0.24	9.9	3.07
XOM	0.266	0.22	10.1	3.13
JPM	0.364	0.96	19.7	2.51
GE	0.330	0.25	9.0	2.65
KO	0.181	0.08	12.8	2.96
PG	0.180	0.11	10.4	2.78
JNJ	0.171	0.23	11.7	2.91
WMT	0.201	0.23	12.9	2.78
INTC	0.327	−0.52	12.3	2.77

Gaussian fit understates the one-percent Value-at-Risk by 68 basis points and the one-percent Expected Shortfall by 208 basis points, more than two percent of a daily return, and it understates the loss in all thirteen series. The kernel, AD-Wiener, and adaptive mixture estimators are within roughly ten basis points of the historical reference at every level. Fig. 16 shows the one-percent Expected Shortfall by series: the Gaussian estimates sit systematically inside the historical loss, while the other three lie on the diagonal. The synthetic kurtotic result of Section XIII is thereby reproduced on real returns; the algebraic and mixture estimators recover the tail risk the Gaussian misses, which is the quantity of regulatory interest.

XV. REAL-DATA HEAPING IN SURVEY SELF-REPORTS

The synthetic heaped experiment of Section VI is borne out on survey microdata from the National Health and Nutrition Examination Survey (NHANES) 2017–2018 [23]. Two complementary studies are reported: a controlled one with a continuous ground truth, and a naturally heaped variable for which no such truth exists.

The body-measures file records each adult’s measured weight, a continuous quantity. Imposing a known heaping grid

TABLE X

TAIL-RISK ACCURACY ON CRSP RETURNS: MEAN ABSOLUTE DEVIATION FROM THE HISTORICAL REFERENCE ACROSS THE THIRTEEN SERIES, IN BASIS POINTS OF DAILY RETURN.

	Gaussian	KDE	AD-Wiener	Mixture
VaR _{1%}	68.1	2.6	4.5	9.6
VaR _{5%}	26.5	2.4	2.1	3.7
ES _{1%}	208.1	8.0	11.1	9.5
ES _{5%}	45.7	2.3	3.1	2.3

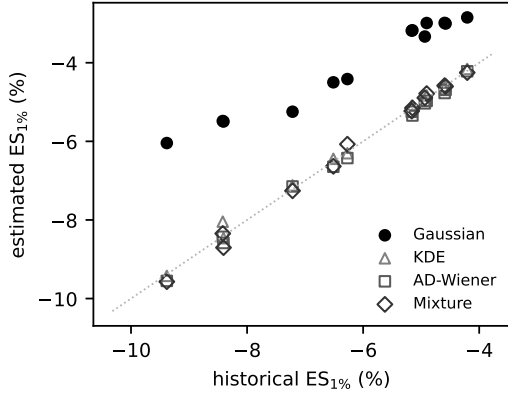


Fig. 16. Estimated against historical one-percent Expected Shortfall for the thirteen CRSP series. The Gaussian estimates lie inside the historical loss (above the diagonal); the kernel, AD-Wiener, and mixture estimates lie on it.

of width D on these measurements and estimating the density from the rounded values isolates the response of each estimator to coarsening, with the no-heaping density, a Gaussian kernel estimate on the unrounded measurements, serving as the reference. This is a controlled robustness study rather than a claim about how weight is reported; self-reported weight in fact heaps mildly, near 5 lb, where every estimator agrees. Table XI gives the integrated squared error against the no-heaping reference as D coarsens. Below about half the kernel bandwidth the heaping is invisible and the three estimators coincide. As D grows the fixed $1/n$ floor is overwhelmed by the rounding spectrum and its error rises by four to five orders of magnitude; the ordinary kernel estimate develops spurious modes at the rounding marks and degrades steadily; the residue floor, read from the spectrum, tracks the elevated level and stays within a small multiple of the no-heaping error throughout. The superposition of Section XI is steadier still: its residual filter reads the rounding comb as structure below the coherence floor and discards it, so the smooth base recovers the de-heaped density at a nearly constant error across the whole range and overtakes the residue floor at the two coarsest grids, where even the residue floor has begun to drift. The price is a slightly higher error at fine heaping, where the residue floor is near exact and the base carries a small parametric bias. Figure 17 shows the divergence and, at a 30 lb grid, the spurious oscillation that the robust estimators suppress.

A variable that heaps coarsely with no imposition is the self-reported number of cigarettes smoked per day, for which the survey instrument itself states that one pack equals twenty

TABLE XI

INTEGRATED SQUARED ERROR ($\times 10^3$) AGAINST THE NO-HEAPING DENSITY AS THE IMPOSED HEAPING GRID D COARSENS, ON NHANES MEASURED ADULT WEIGHTS ($n = 5173$). THE FIXED FLOOR FAILS CATASTROPHICALLY ONCE THE GRID EXCEEDS THE KERNEL BANDWIDTH; THE RESIDUE FLOOR STAYS ROBUST; THE SUPERPOSITION IS FLAT ACROSS THE RANGE, OVERTAKING THE RESIDUE FLOOR AT THE COARSEST GRIDS.

D (lb)	naive KDE	AD simple	AD residue	superposition
5	0.000	0.018	0.006	0.021
10	0.001	0.003	0.006	0.019
20	0.087	306.7	0.009	0.016
30	1.349	501.4	0.037	0.016
40	8.369	718.5	0.087	0.018

cigarettes. Among the 1019 adult smokers reporting a daily count, 42% gave a multiple of ten and 22% a multiple of twenty, against the ten and five percent expected under no digit preference, and the terminal digit zero alone accounts for 42% of reports. This concentration on round numbers reproduces the heaping documented for retrospective cigarette counts [24]. Such counts have no continuous ground truth, since the available objective correlate, serum cotinine, measures exposure and not count, so this study is qualitative. Figure 18 shows the consequence: the ordinary kernel estimate either reproduces the spurious spikes at ten and twenty or, at a bandwidth wide enough to remove them, erases genuine structure, whereas the residue floor reads the heaping as a narrow band of spectral power above the noise level and recovers a smooth density. The residue floor is thus the robust choice precisely where heaping is coarse, the regime into which recall and count data fall.

XVI. REAL-DATA VALIDATION ON PARTICLE-PHYSICS SPECTRA

A second real-data study turns to a setting with no parametric mixture structure: the dimuon invariant-mass spectrum from the CMS experiment [25], in which narrow resonances, the J/ψ at 3.10, the Υ family near 9.5, and the Z at 91.2 GeV, sit on a smooth and steeply falling combinatorial background. Estimating this density is a multi-scale problem: no single bandwidth resolves the narrow peaks and the broad continuum at once, the regime the adaptive estimator targets. Because the underlying density is unknown, the estimators are scored by held-out fit. The 99,165 opposite-sign events are split into a 70 percent training set and a disjoint 30 percent test set; each estimator is built on the training set and scored by the mean negative log-likelihood it assigns to the test set, in the log-mass variable so the decade-spanning resonances are comparably scaled. Table XII reports the result: the adaptive estimators improve on the ordinary kernel estimate by about a quarter of a nat, the fixed bandwidth being unable to sharpen the resonances without roughening the continuum. Figure 19 shows the spectrum and a zoom on the J/ψ region, where the ordinary estimate rounds the peak that the AD-Wiener estimate resolves.

Controlled heaping (measured weight, ground-truthed)

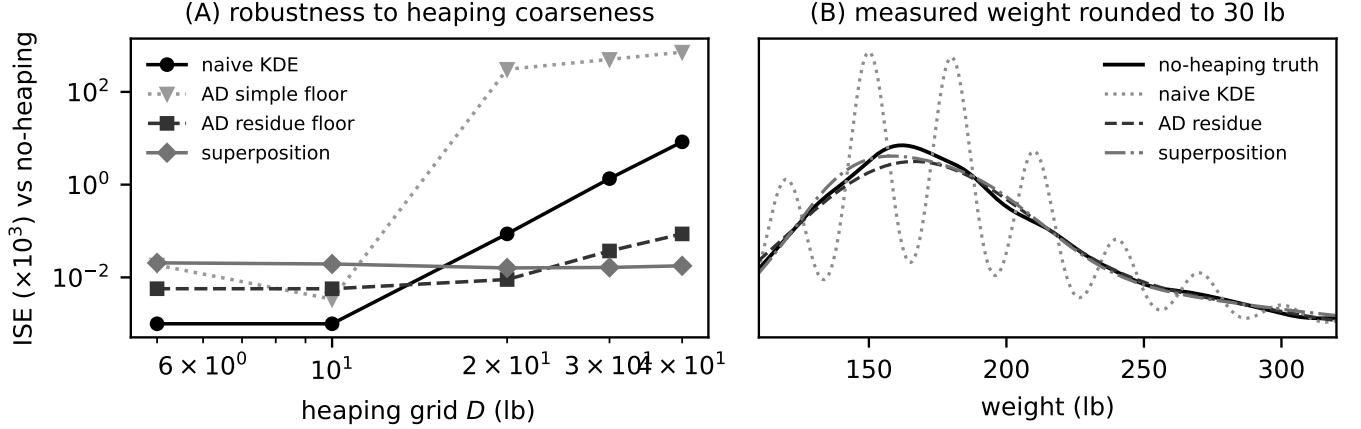


Fig. 17. Controlled heaping on NHANES measured weights. (A) Integrated squared error against the no-heaping density as the imposed grid coarsens: the fixed floor fails catastrophically, the ordinary estimate degrades, the residue floor stays robust, and the superposition is flat across the range, overtaking the residue floor at the coarsest grids. (B) At a 30 lb grid the ordinary estimate oscillates at the rounding marks while the residue floor and the superposition track the truth.

Natural heaping (self-reported cigarettes/day, qualitative)

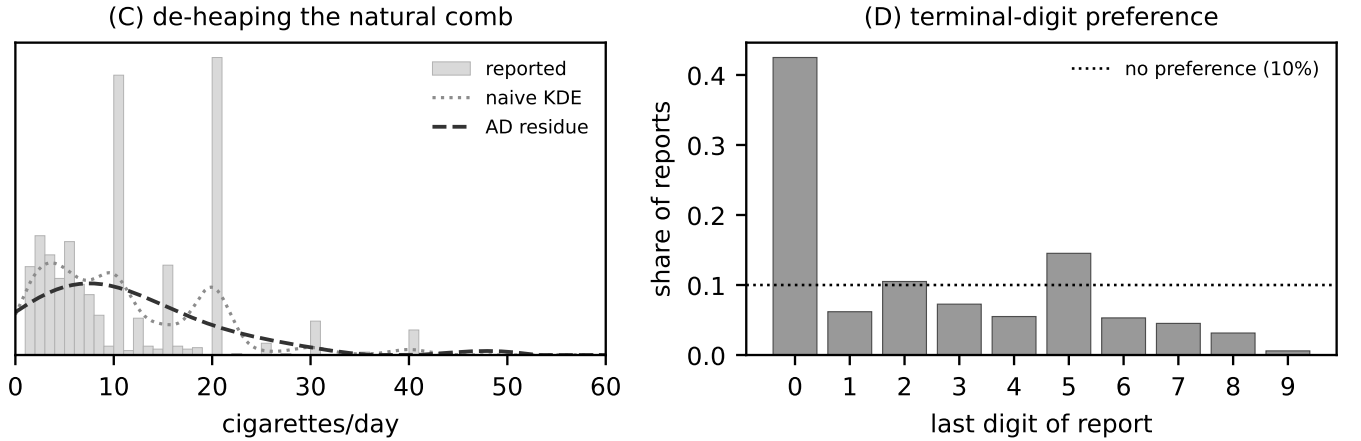


Fig. 18. Natural heaping in self-reported cigarettes per day. (C) The reported counts spike at ten and twenty; the ordinary kernel estimate follows the comb while the residue floor recovers a smooth density. (D) Terminal-digit shares against the ten-percent no-preference line, showing the concentration on zero.

TABLE XII
HELD-OUT MEAN NEGATIVE LOG-LIKELIHOOD (LOG-MASS) ON THE CMS
DIMUON SPECTRUM, LOWER IS BETTER; 99,165 EVENTS, 70/30
TRAIN/TEST SPLIT.

estimator	held-out NLL
naive KDE	0.429
AD-bw	0.315
AD-Wiener	0.314

XVII. REAL-DATA VALIDATION ON GALAXY-REDSHIFT SPECTRA

A further held-out study turns to astronomy, where the same multi-scale geometry arises from an unrelated physical mechanism. In a cone of the sky the spectroscopic galaxy-redshift distribution $n(z)$ is a smooth selection envelope on which large-scale structure, the walls, clusters, and voids of the cosmic web, prints narrow overdensities at particular redshifts. The setting is structurally that of the dimuon spectrum of Section XVI: sharp features carrying real probability mass on a smooth background, a density no single bandwidth

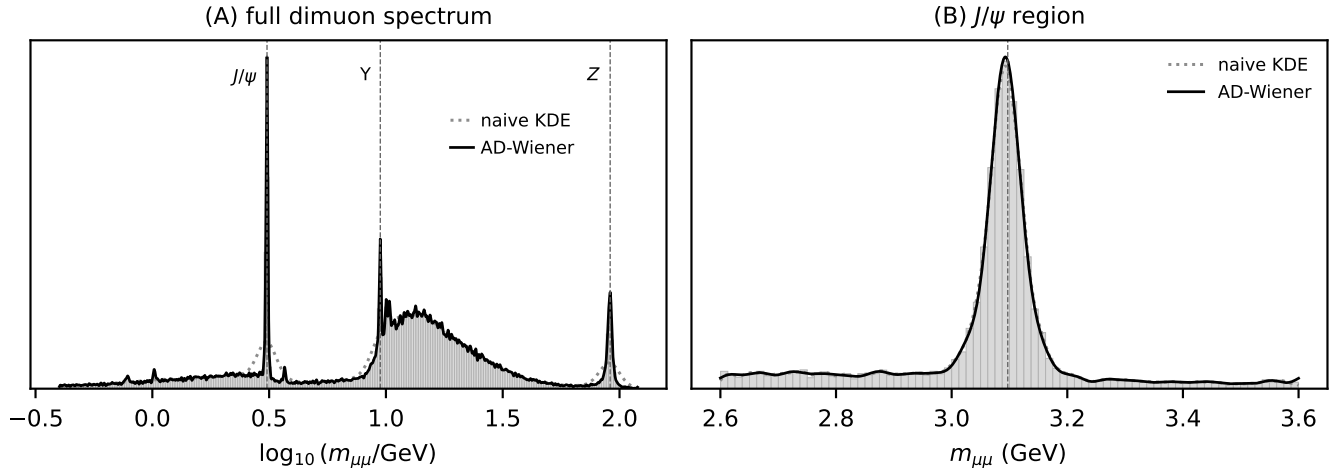


Fig. 19. CMS dimuon invariant-mass spectrum (CMS Open Data, record 545). (A) The full spectrum in log-mass with the J/ψ , Υ , and Z resonances marked; the ordinary kernel estimate rounds the resonances while the AD-Wiener estimate resolves them. (B) The J/ψ region, where the ordinary estimate broadens the peak.

resolves. What it adds is the demonstration that the advantage is statistical rather than physical. The dimuon peaks are quantum resonances of fixed natural width on a combinatorial continuum, whereas the redshift walls are the imprint of galaxy clustering on a survey selection function; the two share nothing physically, yet they present the empirical characteristic function with the same multi-scale profile and the same estimator recovers it. The case thus isolates the claim made throughout, that the adaptive estimator earns its advantage wherever narrow structure carries mass and a global bandwidth is wrong for it, from any particular generative model.

The data are the spectroscopic redshifts of the SDSS DR18 `SpecObj` catalogue [26], retrieved through the SkyServer SQL interface [27]. A five-degree-radius cone about right ascension 200° and declination 0° , restricted to class `GALAXY` with a clean redshift flag and $0.01 \leq z \leq 0.22$, returns 6,684 redshifts. The field is wide enough that the dominant walls, which span many degrees, are sampled by enough galaxies to be resolved rather than lost in counting noise; this sample size is what makes the advantage real, since a narrow pencil beam of a few hundred galaxies does not separate the adaptive gain from resampling variation. As for the dimuon spectrum the true density is unknown, so the estimators are scored by held-out fit on a 70/30 train and test split by the mean negative log-likelihood assigned to the test set.

Table XIII reports the result: both spectral estimators improve on the ordinary kernel estimate by about 0.13 nat, the fixed bandwidth being unable to sharpen the walls without roughening the smooth envelope. The bandwidth selector and the Wiener estimator are indistinguishable here, as they are on the well-sampled dimuon spectrum, so the gain is attributable to the spectral cutoff rather than to the per-frequency taper: the residue floor and its cutoff recover the structure once the data resolve it. Figure 20 shows the $n(z)$ and a zoom on the dominant wall near $z \approx 0.08$, where the ordinary estimate

TABLE XIII
HELD-OUT MEAN NEGATIVE LOG-LIKELIHOOD ON THE SDSS DR18 GALAXY-REDSHIFT DENSITY, LOWER IS BETTER; 6,684 GALAXIES IN A FIVE-DEGREE CONE, 70/30 TRAIN/TEST SPLIT.

estimator	held-out NLL
naive KDE	-1.876
AD-bw	-2.011
AD-Wiener	-2.009

rounds the overdensity that the AD-Wiener estimate resolves.

XVIII. REAL-DATA VALIDATION ON A VERIFIED RANDOM-BEACON STREAM

A fourth completed real-data study moves to a security setting and to a bounded-support target whose true density is known exactly. The source is the NIST randomness beacon [28], which emits a 512-bit random pulse each minute; the study uses 140,000 consecutive pulses, a stream of 71.68×10^6 bits. Before any estimation the stream is checked for corruption by the core tests of the NIST statistical test suite [29], run as in the SMU-DDI STEER framework [30]: the bits are partitioned into sixty-eight sequences of 2^{20} bits, each test is run on each sequence, and the proportion of sequences passing at significance 0.01 is compared to the acceptable interval, $0.99 \pm 3\sqrt{0.99 \cdot 0.01/68}$, lower bound 0.954. Every test lands in the interval (Table XIV), so the stream is taken to be uncorrupted.

The verified bits are read as uniform deviates on $[0, 1)$ by taking successive 32-bit words, so the specified target is the uniform density on a bounded support. The uniform density is exactly the lowest cyclic mode, which AD-Wiener represents at the constant term, so it estimates the target almost exactly (Kullback-Leibler divergence 0.0004, Jensen-Shannon 0.0001 at $N = 8000$). The Gaussian mixture cannot represent

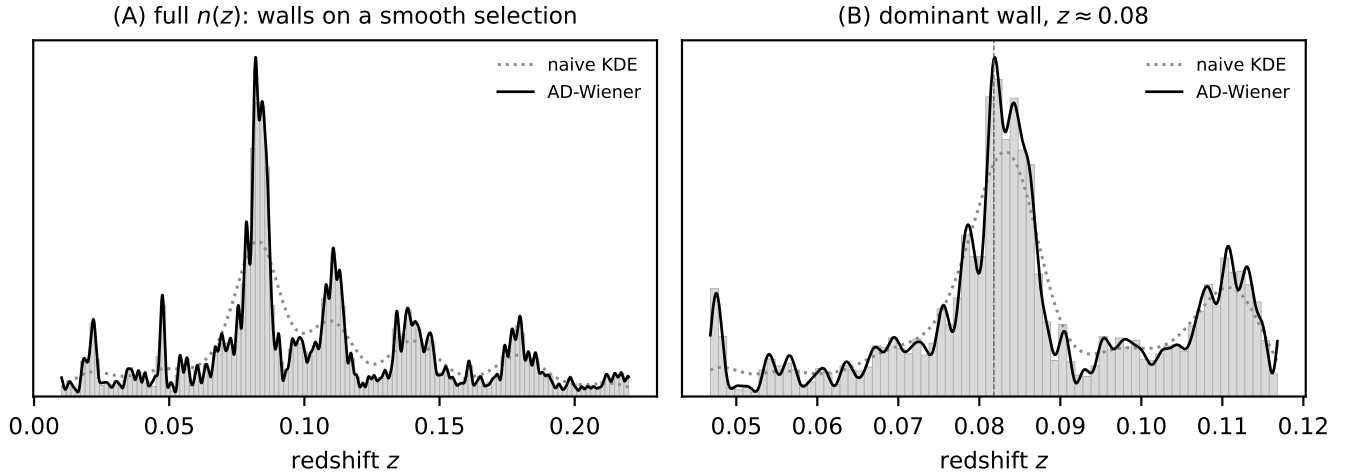


Fig. 20. SDSS DR18 galaxy-redshift density in a five-degree cone about $(\alpha, \delta) = (200^\circ, 0^\circ)$. (A) The full $n(z)$, a smooth selection envelope carrying narrow walls of large-scale structure; the ordinary kernel estimate rounds the walls while the AD-Wiener estimate resolves them. (B) The dominant wall near $z \approx 0.08$, where the ordinary estimate broadens the overdensity that the AD-Wiener estimate resolves.

TABLE XIV

CORE NIST SP 800-22 TESTS ON THE BEACON STREAM, RUN ON SIXTY-EIGHT SEQUENCES OF 2^{20} BITS: NUMBER PASSING AT SIGNIFICANCE 0.01. EVERY PROPORTION IS WITHIN THE ACCEPTABLE INTERVAL (LOWER BOUND 0.954), SO THE STREAM IS UNCORRUPTED.

test	pass/68	test	pass/68
Frequency (monobit)	68	Cumulative sums (fwd)	68
Block frequency	68	Cumulative sums (bwd)	68
Runs	66	Approximate entropy	68
Longest run of ones	65	Serial (1)	67
Spectral (DFT)	66	Serial (2)	68

a flat density with a few components and is an order of magnitude worse (0.0143 / 0.0048), and the ordinary kernel estimate carries the boundary bias of bounded support (0.0069 / 0.0023). The automatic mixed-mode rule of Section X leaves the support entirely on AD-Wiener here, since the mixture wins nowhere, which is the correct outcome for a structureless target and a check that the rule does not invent boundaries where none belong.

An optional uniformity gate sharpens this case to an exact result. Under the uniform null each non-DC squared empirical characteristic coefficient satisfies $n|\hat{c}_k|^2 \sim \text{Exp}(1)$, so its maximum over the K candidate modes concentrates near $\ln K$; the gate declares the stream uniform when that maximum stays below $\ln(K/\alpha)$, an extreme-value threshold at level α whose detectable ripple scales as $1/n$. When it fires, the AD-Wiener estimate is replaced by the exact constant, here taking the Kullback-Leibler divergence to zero; applied to a single AD-Wiener subsection it flattens that stretch alone, joined by (4) so the result stays differentiable and the mass on the interval is preserved. The gate is off by default and is a deliberate prior toward uniformity: it erases structure below the $1/n$ noise floor, so it suits randomness testing and structureless baselining, where any departure from uniform is itself the

signal, rather than general estimation. On the beacon stream it fires and returns the exact uniform; on the structured targets of the earlier sections it does not fire.

XIX. NETWORK-TRAFFIC DENSITY ESTIMATION

A fifth real-data domain is network security, where behavioral baselining models the density of traffic features and the same multi-scale geometry recurs. The data are the labeled flows of the UNSW-NB15 corpus [31] of the Australian Centre for Cyber Security, in the standard training and testing partition; the same approach applies to other corpora such as CIC-IDS2017 [32]. Four continuous flow features are studied on the logarithmic scale: the packet rate, the source load, the destination volume, and the mean destination packet size. Each is heavy tailed with a concentrated benign mode and additional sharp modes contributed by attack traffic, a port scan or a denial-of-service burst collapsing a feature onto a narrow band. The estimators are scored by held-out negative log-likelihood, on a benign-only model and on a benign-plus-attack mixture, with the mixture base the bundled Gaussian-mixture EM as everywhere else in this report.

Table XV reports the result and Fig. 21 shows the mixture density of two features. The superposition attains the lowest held-out negative log-likelihood on every feature and both scenarios, ahead of the global Silverman bandwidth by a wide margin and ahead of the improved Sheather-Jones bandwidth and of AD-Wiener alone; AD-Wiener alone improves on the rule of thumb but not on the best plug-in here. The figure shows why: the benign-plus-attack density carries several sharp modes from distinct traffic and attack types, which the global bandwidth blurs into broad bumps and the adaptive estimator resolves. The advantage holds on benign-only traffic as well as on the attack mixture, so it reflects the intrinsic multi-scale structure of the features rather than the presence of

TABLE XV

UNSW-NB15 HELD-OUT NEGATIVE LOG-LIKELIHOOD (LOWER IS BETTER) ON FOUR LOG-SCALED TRAFFIC FEATURES, ON A BENIGN-ONLY MODEL AND A BENIGN-PLUS-ATTACK MIXTURE; ROW MINIMUM IN BOLD. THE MIXTURE BASE OF THE SUPERPOSITION IS THE BUNDLED GAUSSIAN-MIXTURE EM.

feature	scenario	Silverman	ISJ/Botev	AD-Wiener	superpose
rate	benign	2.199	1.687	1.931	1.613
rate	benign+attack	2.323	1.305	2.163	1.219
sload	benign	2.402	2.073	2.206	2.005
sload	benign+attack	2.558	2.047	2.388	1.940
dload	benign	2.535	1.532	2.238	1.420
dload	benign+attack	2.558	0.571	2.394	0.485
dmean	benign	1.146	-0.671	0.981	-1.372
dmean	benign+attack	1.444	-1.477	1.295	-1.807

attacks, the same situation as the particle-physics and galaxy-redshift spectra of Sections XVI and XVII.

Two qualifications are stated plainly. First, the gain is in density quality, not in detection: scored as a one-class anomaly detector by benign-model density, the adaptive estimators do not improve attack-detection area-under-curve over the global bandwidth, which is comparable or better; a sharper density model is not by itself a better novelty detector, and the contribution here is the traffic density model, not a detector. Second, traffic features are partly discrete and heaped, so the heaping robustness of Section XV carries over, and part of the superposition advantage is its resolution of this systematic structure. The mixture base throughout is the bundled BIC Gaussian-mixture EM, not an external library.

XX. BENCHMARK AGAINST ESTABLISHED ESTIMATORS

The preceding sections set the spectral estimators against ordinary kernel estimation and an adaptive mixture on chosen targets. This section places them on the field's standard external yardstick, the fifteen normal-mixture densities of Marron and Wand [16], whose closed-form densities make the integrated squared error exact rather than estimated. Seven estimators are compared: a Silverman rule-of-thumb kernel; the improved Sheather-Jones plug-in bandwidth of Botev and colleagues [6], the bandwidth of their diffusion estimator and the strongest classical selector here; a true least-squares cross-validation bandwidth (full sample at $n = 100$ and $n = 500$, selected on a thousand-point subsample at $n = 5000$ for tractability); an Abramson adaptive variable-bandwidth kernel; a Gaussian mixture selected by BIC; the AD-Wiener estimator with the residue floor; and the superposition of Section XI with the bundled Gaussian-mixture EM base, written super (GMM). Each density is drawn fifty times at each of three sample sizes, $n = 100$, 500, and 5000, and every estimator is scored by integrated squared error against the true density on a common grid.

Tables XVI, XVII, and XVIII give the per-density errors at the three sample sizes and Fig. 22 the average ranks. The spectral advantage is real but emerges with sample size rather than holding throughout, and this is the central finding of the benchmark. At $n = 5000$ the two spectral estimators, the AD-Wiener filter and the superposition, take the top two average ranks, 2.10 and 2.17, ahead of every classical baseline

including the improved Sheather-Jones bandwidth, with the BIC mixture third at 3.60. At $n = 500$ the superposition leads at 2.57, with AD-Wiener at 3.23 and the cross-validation bandwidth and the BIC mixture a step behind, both at 3.30. At $n = 100$ the order inverts: a plain least-squares cross-validation bandwidth is best at 2.87, AD-Wiener remains the strongest spectral estimator at 3.20, and the superposition falls to mid-field at 4.10, behind the Silverman rule of thumb.

The crossover is a bias-variance tradeoff, and it is worth stating because it delineates where the method helps and where it does not. The AD-Wiener estimator is, by construction, the minimum-mean-squared-error linear filter on the empirical characteristic function: it keeps high-frequency content wherever the data support it and discards it elsewhere, so it introduces little bias and captures structure that a single bandwidth blurs. The price is variance. Estimating a per-frequency filter from the data, rather than committing to one global bandwidth, adds estimation variance, and at small samples that variance dominates. The methods that lead at $n = 100$, the cross-validation bandwidth and the Silverman rule, win precisely by oversmoothing: they trade bias for a large reduction in variance, and at a hundred points that is the right trade, because the problem is variance-limited and there is not enough data to estimate the fine structure reliably, so committing to it adds more variance than the bias it removes. As n grows the variance penalty shrinks while the low-bias advantage persists, the balance tips, and by $n = 500$ the spectral estimators, which were paying for structure they could not yet estimate, estimate it well and take the top ranks. The superposition is the most sample-hungry of the three, because it must also fit a Gaussian-mixture base; at $n = 100$ the base cannot be estimated reliably and the superposition inherits its small-sample variance rather than improving on it, while the band-limited residual has too little signal to separate scales. AD-Wiener alone, which fits no mixture, is therefore the robust spectral choice at small samples, and the superposition is the estimator to use once the sample is large enough to fit its base, which on these densities is by $n = 500$.

At the large sample the division of labor is the one the synthetic battery predicted. The BIC mixture wins the low-order densities whose true form it matches exactly, the Gaussian and the two symmetric bimodals, because the Marron-Wand densities are themselves normal mixtures and the mixture is fitting the correct model class. AD-Wiener wins the genuinely multi-scale densities, the strongly skewed, the trimodal, and the asymmetric double claw. The superposition collects both, matching the mixture on the smooth densities and the band-limited estimator on the sharp ones, and beats the BIC mixture overall even though the benchmark hands the mixture the correct model class, because the residual recovers the multi-scale structure the mixture omits.

Three limitations are stated plainly. The benchmark is a family of normal mixtures, which favors the mixture base; a non-mixture stress test would isolate the spectral contribution further. The comparison set, while it includes the strongest plug-in and cross-validation bandwidths and an adaptive kernel, omits the log-spline and taut-string estimators, which are also strong on multi-scale densities and remain to be added.

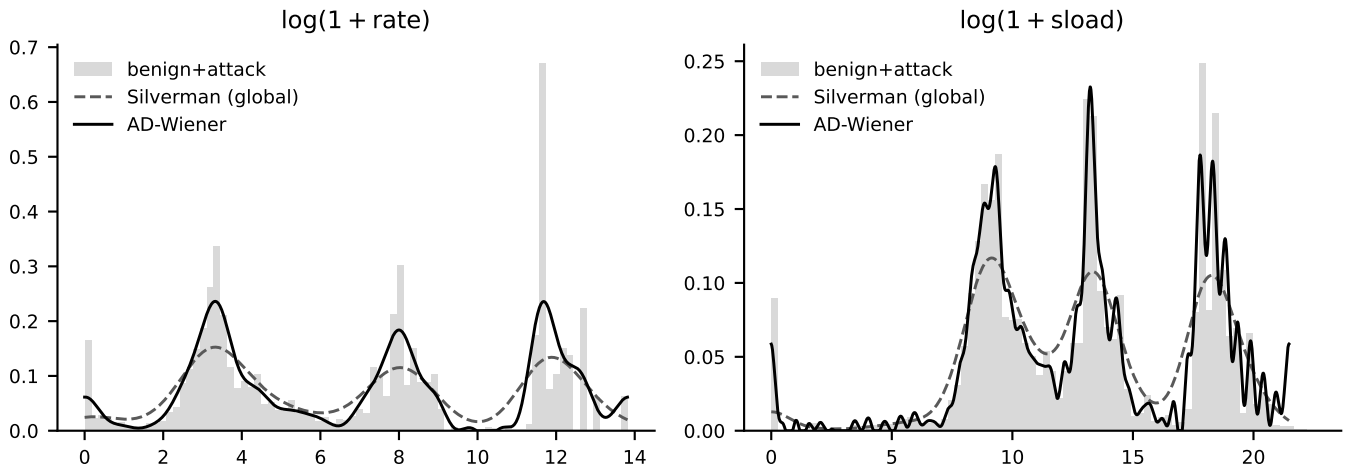


Fig. 21. UNSW-NB15 benign-plus-attack density of the packet rate and source load on the logarithmic scale. The global Silverman bandwidth (dashed) blurs the several traffic modes into broad bumps; the adaptive AD-Wiener estimator (solid) resolves them.

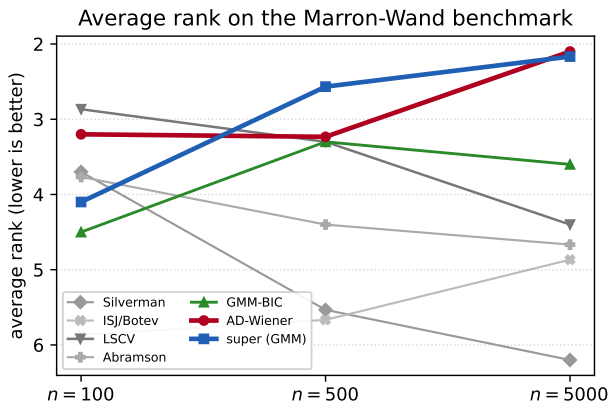


Fig. 22. Average rank over the fifteen Marron-Wand densities, lower is better, at $n = 100$, 500 , and 5000 . The spectral advantage emerges with sample size: at $n = 5000$ AD-Wiener and the superposition take the top two ranks ahead of every classical baseline, while at $n = 100$ a cross-validation bandwidth leads and the superposition trails.

And the superposition is not the estimator of choice at small samples: below roughly $n = 500$ on these densities a cross-validation bandwidth or AD-Wiener alone is preferable, since the mixture base needs data to fit. At moderate and large samples, within the set tested, the superposition attains the best average rank among established estimators.

XXI. SOFTWARE, REPRODUCIBILITY AND DATA AVAILABILITY

The estimators and experiments are released as a reproducible bundle, available at <https://github.com/mitch-thornton/kde-ad-wiener>. The mixture and group extensions are organized behind a small plugin interface with three hooks: a candidate group library, a blind group-matching backend, and an adaptive mixture estimator. Each ships a self-contained default, the cyclic group, the cyclic matched group, and

a Gaussian-mixture estimator with model-selected order, so every figure and table reproduces with no external dependency. Every mixture result reported here uses this bundled BIC Gaussian-mixture EM. A practitioner may register a richer group library, a blind group-matching engine, or an alternative mixture estimator in place of the defaults. The real-data study of Section XIV is likewise scripted end to end: an extraction step writes the CRSP series into a `data/` directory and an analysis step writes the risk, Value-at-Risk, and tail-risk results into a `results/` directory as JSON, separating the licensed-data step from the dependency-free analysis. The heaping study of Section XV is scripted the same way against the public NHANES files, which the analysis step reads directly to regenerate both figures. The particle-physics and galaxy-redshift studies of Sections XVI and XVII are scripted likewise, each reading a public released or exported CSV directly to regenerate its figure and held-out table.

The CRSP daily stock data of Section XIV are proprietary and available to subscribers through Wharton Research Data Services [33]; the extraction script and the derived per-series results that reproduce the figures are included in the bundle. The NHANES 2017–2018 files of Section XV are public [23]; the analysis script regenerates both figures from them. The CMS dimuon data of Section XVI are public (CERN Open Data, CC0) [25]; the analysis script reads the released CSV directly. The SDSS DR18 galaxy redshifts of Section XVII are public [26], retrieved through the SkyServer SQL interface [27]; the analysis script reads the exported CSV directly.

XXII. CONCLUSION

Kernel density estimation is recast by reading bandwidth selection as a spectral-support problem in the characteristic-function domain: the binned data’s group-averaged spectrum is the squared empirical characteristic function, and the bandwidth is the cutoff where it meets the $1/n$ floor. The resulting selector matches the rule of thumb on smooth densities and approaches the best fixed bandwidth on hard ones. Going

TABLE XVI

INTEGRATED SQUARED ERROR ($\times 10^3$) ON THE FIFTEEN MARRON-WAND DENSITIES AT $n = 100$, MEAN OVER FIFTY REPLICATIONS; LOWER IS BETTER, ROW MINIMUM IN BOLD. GMM-BIC IS A GAUSSIAN MIXTURE SELECTED BY BIC; SUPER (GMM) IS THE SUPERPOSITION OF SECTION XI WITH THAT MIXTURE AS ITS SMOOTH BASE.

density	Silverman	ISJ/Botev	LSCV	Abramson	GMM-BIC	AD-Wiener	super (GMM)
Gaussian	5.83	18.36	10.53	7.24	2.64	9.98	2.64
Skewed unimodal	9.05	29.52	13.38	9.57	12.18	11.15	13.30
Strongly skewed	142.2	86.35	48.64	112.5	87.07	53.24	53.45
Kurtotic unimodal	101.7	97.43	52.40	58.83	79.23	55.80	56.22
Outlier	61.34	313.5	85.01	71.00	45.32	56.71	31.90
Bimodal	8.73	21.18	11.02	8.68	16.96	12.46	17.54
Separated bimodal	46.74	32.59	14.94	40.26	9.12	16.15	16.34
Skewed bimodal	12.57	22.48	12.99	12.75	22.89	13.61	21.21
Trimodal	11.16	20.63	12.69	11.33	16.47	13.59	16.11
Claw	53.17	44.85	42.47	54.59	57.88	45.19	55.26
Double claw	10.21	22.44	12.10	10.35	20.10	13.56	18.70
Asymmetric claw	27.81	31.55	28.18	28.39	33.50	26.30	31.43
Asym. double claw	14.42	25.66	17.54	14.07	21.44	17.68	21.92
Smooth comb	85.06	51.44	39.10	79.85	45.18	41.65	41.91
Discrete comb	113.5	51.56	39.15	113.5	39.55	40.69	40.55
avg. rank	3.70	5.87	2.87	3.77	4.50	3.20	4.10
wins (of 15)	4	0	5	2	2	1	1

TABLE XVII

INTEGRATED SQUARED ERROR ($\times 10^3$) ON THE FIFTEEN MARRON-WAND DENSITIES AT $n = 500$, MEAN OVER FIFTY REPLICATIONS; LOWER IS BETTER, ROW MINIMUM IN BOLD. COLUMNS AS IN TABLE XVI.

density	Silverman	ISJ/Botev	LSCV	Abramson	GMM-BIC	AD-Wiener	super (GMM)
Gaussian	1.86	4.73	2.54	2.24	0.48	2.37	0.48
Skewed unimodal	2.85	7.52	4.14	2.60	2.33	2.74	1.86
Strongly skewed	106.3	30.09	15.69	73.70	42.54	16.71	16.59
Kurtotic unimodal	56.23	30.20	13.96	16.31	8.28	12.47	12.61
Outlier	19.62	59.86	23.53	21.54	6.17	15.45	6.15
Bimodal	3.01	6.05	3.16	2.19	1.61	3.16	1.84
Separated bimodal	20.86	8.76	3.97	13.02	1.66	3.69	3.75
Skewed bimodal	5.20	7.10	4.09	4.03	7.76	3.73	3.86
Trimodal	4.81	7.05	3.62	3.71	5.42	3.96	5.67
Claw	45.95	25.03	12.49	46.03	49.20	14.21	42.47
Double claw	4.43	7.26	4.32	3.60	3.07	4.52	3.32
Asymmetric claw	20.82	14.41	11.03	20.12	20.49	11.70	19.71
Asym. double claw	7.87	8.61	7.28	6.76	6.08	7.42	6.23
Smooth comb	63.54	20.89	16.40	58.54	20.36	17.43	17.44
Discrete comb	88.28	19.43	16.06	83.11	22.44	16.43	16.20
avg. rank	5.53	5.67	3.30	4.40	3.30	3.23	2.57
wins (of 15)	0	0	6	0	6	1	2

beyond a fixed kernel, the per-frequency optimal taper gives an adaptive estimator that surpasses the best fixed bandwidth on most standard densities, including sharply peaked and comb-like cases on which fixed-bandwidth methods fail, and it extends in the same domain to deconvolution under known measurement error. Because that estimator resolves sharp structure but does not fit a smooth base as economically as a mixture, a Gaussian mixture is combined with it as a piecewise partition and as a superposition of a smooth base and a band-limited residual, the superposition being made the default. On the standard Marron-Wand benchmark, scored by exact integrated squared error across three sample sizes, the advantage emerges with sample size through a bias-variance tradeoff: the spectral estimators carry low bias but pay in variance, so a cross-validation bandwidth leads at $n = 100$ and AD-Wiener alone is the robust choice, while at $n = 5000$ the Wiener filter and the superposition take the top two average ranks ahead of every classical baseline including the Botev diffusion

bandwidth, the superposition beating even a Gaussian mixture handed the correct model class. The same spectrum supplies a data-driven floor robust to heaped data, and the estimators are validated on six real datasets, recovering tail risk a Gaussian fit understates on CRSP returns, staying robust to rounding on NHANES self-reports, improving held-out likelihood on the multi-scale CMS dimuon and SDSS galaxy-redshift spectra, recovering the uniform target of a verified NIST randomness-beacon stream almost exactly, and giving the lowest held-out likelihood on UNSW-NB15 network-traffic features, with a synthetic-data quality-checking use-case recovering contrived target densities. All claims are validated numerically and the experiments are reproducible.

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TABLE XVIII

INTEGRATED SQUARED ERROR ($\times 10^3$) ON THE FIFTEEN MARRON-WAND DENSITIES AT $n = 5000$, MEAN OVER FIFTY REPLICATIONS; LOWER IS BETTER, ROW MINIMUM IN BOLD. COLUMNS AS IN TABLE XVI.

density	Silverman	ISJ/Botev	LSCV	Abramson	GMM-BIC	AD-Wiener	super (GMM)
Gaussian	0.31	0.69	0.47	0.37	0.06	0.28	0.06
Skewed unimodal	0.48	1.12	0.72	0.51	0.46	0.37	0.24
Strongly skewed	58.90	5.04	3.96	30.74	9.42	2.32	2.32
Kurtotic unimodal	18.41	4.43	3.04	1.88	2.12	1.63	1.63
Outlier	2.89	7.29	4.17	3.28	0.62	1.55	0.62
Bimodal	0.60	0.94	0.61	0.38	0.15	0.38	0.16
Separated bimodal	5.07	1.35	0.83	1.49	0.15	0.46	0.15
Skewed bimodal	1.18	1.16	0.82	0.49	1.29	0.44	0.18
Trimodal	1.35	1.16	0.89	0.75	1.28	0.53	1.29
Claw	32.93	3.99	2.69	26.42	2.35	1.50	1.50
Double claw	2.09	2.07	2.06	1.85	1.66	1.85	1.66
Asymmetric claw	12.51	3.32	3.49	10.73	7.76	2.57	6.17
Asym. double claw	4.74	2.59	2.77	4.23	4.58	2.32	4.59
Smooth comb	40.79	5.46	7.57	37.55	11.24	3.80	3.78
Discrete comb	47.65	5.39	7.46	39.19	14.92	2.43	2.42
avg. rank	6.20	4.87	4.40	4.67	3.60	2.10	2.17
wins (of 15)	0	0	0	0	5	6	4

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